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Motion of a scalar particle in the environment of gravitating scalar field

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Abstract: We derive the equations of motion for FJNW metric and its ultrastatic limit in the presence of a scalar field for a massive scalar test particle using the Hamilton's principle and d'Alembert's principle. We then compare these equations with the standard geodesic equations in their corresponding spacetimes. Additionally we compare it to the special case when the reparametrized FJNW metric becomes the exterior of Schwarzschild's metric and the possibility of circular orbits. Furthermore, we explore the effective potential of these spacetimes with different particle charges.

Keywords: FJNW metric, naked singularities, scalar field, ultrastatic limit, variational principle

Název práce: Pohyb skalární částice v prostředí gravitujícího skalárního pole

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Abstrakt: V této práci odvodíme rovnice pohybu pro FJNW metriku a její ultrastatickou limitu v přítomnosti skalárního pole pro hmotnou skalární testovací částici s použitím Hamiltonova principu a d'Alembertova principu. Tyto rovnice poté porovnáme s rovnicemi vycházející ze standardního geodetického pohybu v odpovídajících prostoročasech. Kromě toho porovnáme speciální případ kdy je FJNW metrika identická s vnějším řešením Schwarzschildovy metriky a prozkoumáme možnosti pro kruhové orbity v těchto prostoročasech. Nakonec prozkoumáme efektivní potenciál v těchto prostoročasech s různým nábojem částice.

Klíčová slova: FJNW metrika, nahé singularity, skalární pole, ultrastatická limita, variační princip

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Introduction

One of the most, if not the most significant scalar fields in physics is the Higgs field, whose excitation produces the well-known Higgs boson. The Higgs boson was hypothesized to exist in the 1960s by François Eglert, Robert Brout [1], Peter Higgs[2], Gerald Guralnik, Thomas Kibble and Carol Richard Hagen[3]. Its existence was confirmed following a series of measurements at the Large Hadron Collider at CERN, as announced in March 2013. The Higgs field itself is a scalar field that interacts with elementary particles through the Higgs mechanism, which in turn gives particles their mass. It is hypothesized, that after the Big Bang, the forces were unified until the Higgs field emerged, which is estimated at a time $t \approx 10^{-11}$s. The Higgs boson has been measured during the ATLAS and CMS experiments[4] to have a fairly heavy mass of $m_H = 125.09 \pm 0.24$GeV. However the most recent measurement has been made by ATLAS[5] $m_H = 125.11 \pm 0.11$GeV on July in 2023.

The Einstein field equations permit naked singularities - curvature singularities not covered by a black hole's event horizon[6]. Naked singularities remain a somewhat controversial topic in modern physics, since they could, in principle, allow any amount of matter or radiation to be expelled form their region of spacetime. That would violate the predictability, or in a sense determinism, which has long been considered one of the fundamental assumptions in this area of physics.

An example of these solutions constitutes the Fisher-Janis-Newman-Winicour solution. The solution, first discovered by Fisher[7] in 1947 and later by Janis, Newman and Winicour[8] in 1968, is now called FJNW solution, which is a spherically symmetric spacetime with a scalar field, that possesses a naked singularity. Wyman also discovered a spherically symmetric solution[9] in 1981, but it has been shown[10] that these solutions are the same.

The Weak Cosmic censorship hypothesis forbids the existence of a naked singularity, i.e. any space-time singularity arising in a collapse from which a signal can originate which escapes to infinity[11]. The strong version of this hypothesis admits only singularities resulting from the same gravitational collapse that are invisible to all observers.

Roger Penrose in his work does not consider the Big Bang to be a naked singularity, since he defines a singularity to be naked if there is an observer (a time-like curve) for whom the singularity lies initially to his future and subsequently to his past. However, in the current understanding, the Big Bang is taken as a space-like naked singularity. We can therefore treat it as a motivation to explore this subject in more depth.

Since we are dealing with naked singularities, Penrose diagrams are a useful visualization of the causal structure of the given system. For a complete understanding, we need to distinguish between various types of horizons, where we use the classification defined in[12]. Future horizon is a boundary of the causal past of $\mathscr{I}^+$. Similarly, the past horizon is a boundary of the causal future of $\mathscr{I}^-$. The event horizon in[12] is defined as the causal boundary. It is a null hypersurface spanned by null geodesics.

However, Penrose defined the absolute event horizon in[11] in the following way. Given the set $\mathcal{E}$ such that it contains all events from which a time-like, respectively

a null curve can be drawn into the future to i^+ and $\mathscr{I}^+$ respectively. Then the boundary of this set $\partial\mathcal{E}$ is the absolute event horizon.
If there exists a time-like Killing vector field in a certain region of spacetime, the Killing horizon can be defined as a hypersurface on which that field is null[12].
The Cauchy problem is an initial value problem of a system whose evolution is uniquely and completely determined by its initial (boundary) conditions. The Cauchy horizon is a future boundary of the Cauchy problem, that is, the boundary of a region where the evolution remains determined by the initial data. A sufficient but not necessary condition for a Cauchy horizon to appear is the existence of a time-like singularity in the given spacetime[12].
Naked singularities are present in different spacetimes with varying degrees of significance, some of which are considered unphysical by their nature. We know that in the 'normal' Schwarzschild case, there is no naked singularity. However if we posit the mass negative, $m < 0$, then there can also emerge a naked singularity in the Schwarzschild solution[13]. It is important to note that this is (for now) a purely theoretical aspect of this solution, since there is no obvious physical interpretation of such a condition. However for the rotating black hole, described using the Kerr metric, and imposing only physical parameters, there also emerges a naked ring-shaped singularity at $r = 0$ if the rotation parameter fulfils the condition $a > M$. And in the Reissner-Nordström solution with $Q^2 < M^2$ remains somewhat ambiguous whether or not the singularity $r = 0$ is truly naked. In principle, there is a singularity that can be seen by a time-like observer only if we extend the spacetime beyond the Cauchy horizon. However, even though the solution can be extended beyond the Cauchy horizon, it has been shown that the Cauchy horizon is unstable and therefore can still be regarded as a boundary of spacetime[13]. And in the extreme case of $e^2 > m^2$, the singularity becomes globally naked[13].
Moreover, singularities in these solutions have a different space-like character. In the Schwarzschild black hole, there is a space-like singularity, meaning it lies within the future or past light-cones for all observers inside a given region, i.e., the event horizon in this case. Null singularities are singularities on null surfaces, which we will precisely study using the FJNW metric. And for the last case, time-like singularities are present in the Kerr and Reissner-Nordström solutions.
A curvature singularity is usually defined as a region in spacetime, where the curvature scalars diverge. This also means that the geodesics are incomplete in these spacetimes, which in addition has the issue, that a time-like observer can cease to exist in finite time[14].
Since we will study spherically symmetric spacetimes, we will also try to look at circular orbits in all cases. Now, noting that these need not exist for all cases, there should exist stable and unstable circular orbits for the FJNW spacetime in different cases based on the parameter defining the strength of the scalar field and particle's mass and scalar charge[15].

1 Lagrangian of a particle in a massless scalar field

Throughout this thesis we will be using Einstein summation convention. We will also adopt the usual general metric g, corresponding to $g^{\mu\nu}$, respectively $g_{\mu\nu}$, with which we will be raising, respectively lowering indices. If explicit use of flat spacetime is considered, this metric will reduce to $\eta_{\mu\nu}$.
The sign convention used here will be $(-,+,+,+)$, unless stated otherwise.
Furthermore we will adhere to the standard notation, where Greek indices range from 0-3 and Latin indices range from 1-3.
Unless explicitly stated, we will assume that all quantities are functions of the spacetime coordinates x^μ.

1.1 Constructing the Lagrangian $\mathcal{L}$

The Lagrangian $\mathcal{L}$ for any system must be invariant and dependent only on the variables describing the system's behaviour.
The whole approach to finding the correct Lagrangian is guided by Occam's razor, which states that we should minimize the number of variables and their complexities in our equations while still capturing the essential physics of the system.
A massive scalar particle on a curved spacetime implies a possible dependency on variables such as the mass of the particle, its (scalar) charge, coordinates of the particle, its four-velocity, potential of the scalar field, and its gradient.
We therefore postulate that $\mathcal{L}$ is at most a function of

$$\mathcal{L} = \mathcal{L}(m_0, q, x^\mu, u^\mu, \phi, \phi_{,\nu}), \tag{1.1}$$

where m_0 is the rest mass of the particle, x^μ is a spacetime coordinate, u^μ is the four-velocity, $\phi(x^\mu)$ the value of a given scalar field at x^μ and q is the scalar charge of the particle.
We assume the scalar charge of a particle to be a given inherent property of the particle, which is invariant. This property is closely related to the law of conservation of charge, which we will **postulate**.
Let us now try to deduce the form of the Lagrangian given by our defined problem and compare it with the results from [16]. We will start only from the coordinate dependence of the Lagrangian and our definition of force.

1.1.1 Field dependent mass

First, we derive the identity which shows a dependence of mass on the scalar field potential. Let us then write out the scalar product of the four-force with four-velocity in flat coordinates, i.e. $g_{\mu\nu} = \eta_{\mu\nu}$. We will now make the very important ansatz, such that $p^\mu = mu^\mu$, where m is not the usual rest mass, but depends also on the value of the scalar field at the particle's position. We will explain the reason behind this later on, together with physical derivation of the

fact. Nevertheless, considering the scalar product, we have

$$\eta_{\mu\nu}F^{\mu}u^{\nu} = \eta_{\mu\nu}\frac{\mathrm{d}(mu^{\mu})}{\mathrm{d}\tau}u^{\nu} = \eta_{\mu\nu}\frac{\mathrm{d}m}{\mathrm{d}\tau}u^{\mu}u^{\nu} + \eta_{\mu\nu}ma^{\mu}u^{\nu} =$$
$$-c^2\frac{\mathrm{d}m}{\mathrm{d}\tau} + \eta_{\mu\nu}ma^{\mu}u^{\nu} = -c^2\frac{\mathrm{d}m}{\mathrm{d}\tau},$$

where in the third equality we have used the relation for normalization of four-velocity on its world-line and in the last one the fact that the four-acceleration is orthogonal to four-velocity.
We will now generalize this equations and rewrite it in curved spacetime - transforming the total derivative to the absolute one

$$g_{\mu\nu}F^{\mu}u^{\nu} = g_{\mu\nu}\frac{D(mu^{\mu})}{d\tau}u^{\nu} = g_{\mu\nu}\frac{\mathrm{d}m}{\mathrm{d}\tau}u^{\mu}u^{\nu} + g_{\mu\nu}m\frac{Du^{\mu}}{d\tau}u^{\nu} =$$
$$-c^2\frac{\mathrm{d}m}{\mathrm{d}\tau} + \frac{1}{2}(\frac{D}{d\tau}(g_{\mu\nu}u^{\mu}u^{\nu}) - \frac{D(g_{\mu\nu})}{d\tau}u^{\mu}u^{\nu}) = -c^2\frac{\mathrm{d}m}{\mathrm{d}\tau},$$

Where we have used the fact, that the metric is symmetric, constant with respect to the covariant derivative, and that the absolute derivative of the term $g_{\mu\nu}u^{\mu}u^{\nu}$ is zero.
We will now make the following **ansatz**

$$F^{\mu} = -q\phi^{,\mu}. \tag{1.2}$$

The minus sign in front of the term on the right corresponds to a physical solution of the problem, as will be shown in the following text. However the sign is purely our choice that is without loss of generality, since it can be absorbed using a redefined charge q.
Furthermore we can use our definition of the force

$$\eta_{\mu\nu}F^{\mu}u^{\nu} = -\eta_{\mu\nu}q\phi^{,\mu}u^{\nu} = -q\phi^{,\mu}u_{\mu} = -q\frac{\mathrm{d}\phi}{\mathrm{d}\tau}.$$

However this generalizes easily to a curved background, since ϕ is a scalar.
This, with comparison to the previous result, gives

$$c^2\frac{\mathrm{d}m}{\mathrm{d}\tau} = q\frac{\mathrm{d}\phi}{\mathrm{d}\tau}.$$

From here, we can state that $m = m(\phi(x^{\mu}))$, as mentioned in our ansatz. After integration over τ and explicitly writing out the coordinates, we get to the relation

$$m(\phi(x^{\mu})) = q\phi(x^{\mu}) + k, \tag{1.3}$$

where k is a scalar constant (since ϕ and m are both scalar variables) independent of time and it has the same units as m.
This relation states that the mass m is dependent on the scalar potential at the particle's position - an important result.
We can extend this result and state, that, since k is constant everywhere, we set it to be the usual rest mass, which is denoted as m_0. This is the mass the particle would posses when the scalar field is zero.

We can then split the rest mass m into two parts: one, which is constant, and the other, which varies with the particles position. We can write the mass as

$$m(x^\mu) = q\phi(x^\mu) \pm m_0,$$

where we are still unsure about the sign in front of the constant mass.
Note that from a purely physical aspect, the sign in front of the constant mass should be positive, so that we do not end up with negative mass if we consider a particle with no charge. If we take this into account, then this simply becomes

$$\boxed{m(x^\mu) = q\phi(x^\mu) + m_0}. \tag{1.4}$$

This then ensures that the mass of the particle remains non-negative in the special case of no scalar field, and thus physical. This form of the mass dependence is therefore physically reasonable and consistent with our expectations.
The Lagrangian given in (1.1) will be greatly simplified from this result, and, since $\phi = \phi(x^\mu)$ only, then, symbolically,

$$\mathcal{L} = \mathcal{L}(x^\mu). \tag{1.5}$$

So the Lagrangian depends only on the coordinates.

1.1.2 Euler-Lagrange equations

From [16] we know, that if the self-forces of a particle acting on itself are omitted, the action (with convention (+,-,-,-)) in flat spacetime is given by

$$S[x(s)] = -\int ds(m_0 + q\phi(x(s))\sqrt{\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}. \tag{1.6}$$

Here, the convention $\dfrac{\mathrm{d}a}{\mathrm{d}s} = \dot{a}$ is used. We will also adopt this convention when considering proper time τ, i.e. $\dfrac{\mathrm{d}a}{\mathrm{d}\tau} = \dot{a}$. When the parameter s, over which we integrate, is the proper time (from here on denoted as τ), then the term in the square root is identically one (-1 for our convention).
Let us then infer from this result that we have the following Lagrangian

$$\mathcal{L} = -(m_0 + q\phi), \tag{1.7}$$

noting that the parameter s is characterized by the proper time $s = \tau$. Note that this complies with our result stating the coordinate dependence of the Lagrangian(1.5).
For the equations of motion, we can use Euler-Lagrange equations given by

$$\frac{\partial\mathcal{L}}{\partial x^\nu} - \frac{\mathrm{d}}{\mathrm{d}\tau}\left[\frac{\partial\mathcal{L}}{\partial u^\nu} + \frac{1}{c^2}\left(\frac{\partial\mathcal{L}}{\partial u^\nu}u^\nu - \mathcal{L}\right)u_\nu\right] = 0, \tag{1.8}$$

which are then reduced by assuming only the coordinate dependency of the Lagrangian to

$$\frac{\partial\mathcal{L}}{\partial x^\nu} + \frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{1}{c^2}\mathcal{L}u_\nu\right) = 0. \tag{1.9}$$

We can then substitute into our Lagrangian from (1.7)

$$(m_0 + q\phi)_{,\nu} + \frac{1}{c^2}\frac{\mathrm{d}}{\mathrm{d}\tau}\left[(m_0 + q\phi)u_\nu\right] = 0. \tag{1.10}$$

In total, we can write

$$(m_0 + q\phi)_{,\nu} + \frac{1}{c^2}(m_0 + q\phi)a_\nu + \frac{1}{c^2}u_\nu\frac{\mathrm{d}}{\mathrm{d}\tau}(m_0 + q\phi) = 0. \tag{1.11}$$

Note, that this equation is still valid in curved spacetime, since the absolute and covariant derivatives of scalars are the same as the usual ones, and the absolute derivative of the four-velocity is the four-acceleration. We can rewrite this using our mass in (1.4) to get the following relation using natural units

$$m_{,\nu} + ma_\nu + u_\nu\frac{\mathrm{d}}{\mathrm{d}\tau}m = 0. \tag{1.12}$$

This is equivalent to

$$\frac{D(mu_\nu)}{d\tau} = -m_{,\nu}, \tag{1.13}$$

which will be important and seen once more later on.

1.1.3 Variational principles

We will be using the d'Alembert's principle with Hamilton's principle. After integrating over proper time, d'Alembert's principle yields

$$\int_{\tau_1}^{\tau_2}\left(F_\mu - \frac{Dp_\mu}{d\tau}\right)\delta x^\mu d\tau = 0, \tag{1.14}$$

where δx^μ is an infinitesimal virtual displacement and F_μ is a combination of external forces. However we will do this calculation in a flat spacetime and then generalize the results.
We will be considering a fixed-end problem to make the final relation simpler and the calculations feasible.
First, we need to prepare the relations which we will be using. For this section, we will follow a very similar calculation as in[17]. Let us start with a variation of proper time $\delta d\tau$. Variation will be denoted by the substitution of $\hat{x}^\mu \to x^\mu + \delta x^\mu$, where $\delta dx^\mu = d\hat{x}^\mu - dx^\mu$. During this variation we need to account also for the variation of proper time, which is not the same for a different world-line. We will need the equation for the proper time on the varied world-line $d\hat{\tau}$

$$\begin{aligned} d\hat{\tau} = \frac{1}{c}\sqrt{-\eta_{\mu\nu}d\hat{x}^\mu d\hat{x}^\nu} = \frac{1}{c}\sqrt{-\eta_{\mu\nu}(dx^\mu + \delta dx^\mu)(dx^\nu + \delta dx^\nu)} = \\ \frac{1}{c}\sqrt{-\eta_{\mu\nu}dx^\mu dx^\nu}\sqrt{1 + 2\frac{\eta_{\mu\nu}dx^\mu\delta dx^\nu}{\eta_{\mu\nu}dx^\mu dx^\nu} + O((\delta x^\mu)^2)} \approx \\ \frac{1}{c}\sqrt{-\eta_{\mu\nu}dx^\mu dx^\nu} + \frac{1}{c}\frac{\eta_{\mu\nu}dx^\mu\delta dx^\nu}{\eta_{\mu\nu}dx^\mu dx^\nu}\sqrt{-\eta_{\mu\nu}dx^\mu dx^\nu} = \\ d\tau + \frac{1}{c}\frac{\eta_{\mu\nu}dx^\mu\delta dx^\nu}{-\sqrt{-\eta_{\mu\nu}dx^\mu dx^\nu}} = \\ d\tau + \frac{1}{c^2}\frac{\eta_{\mu\nu}dx^\mu\delta dx^\nu}{-d\tau}, \end{aligned}$$

where we have neglected the terms of order $(\delta x^\mu)^2$ and used the approximation $\sqrt{1+\epsilon} = 1 + \frac{1}{2}\epsilon + O(\epsilon^2)$.
In total, we have

$$\delta d\tau = \frac{\eta_{\mu\nu} dx^\mu \delta dx^\nu}{-c^2 d\tau}, \tag{1.15}$$

from where we can express (after renaming the indices) the term δdx^μ as

$$\delta dx^\mu = -\frac{c^2 d\tau \delta d\tau}{\eta_{\mu\nu} dx^\nu} = -\frac{c^2 d\tau \delta d\tau}{dx^\mu}. \tag{1.16}$$

Therefore we can rewrite the d'Alembert principle

$$\begin{aligned}
\int_{\tau_1}^{\tau_2} \left(F_\mu - \frac{\mathrm{d}p_\mu}{\mathrm{d}\tau}\right) \delta x^\mu d\tau = \int_{\tau_1}^{\tau_2} \left(F_\mu - \frac{\mathrm{d}p_\mu}{\mathrm{d}\tau}\right) \delta x^\mu d\tau \overset{P.P.}{=} \\
\cancelto{0}{[-p_\mu \delta x^\mu d\tau]_{\tau_1}^{\tau_2}} + \int_{\tau_1}^{\tau_2} \left(F_\mu \delta x^\mu + p_\mu \frac{\mathrm{d}\delta x^\mu}{\mathrm{d}\tau}\right) d\tau = \\
\int_{\tau_1}^{\tau_2} (F_\mu \delta x^\mu d\tau + p_\mu d\delta x^\mu) = \\
\int_{\tau_1}^{\tau_2} \left(F_\mu \delta x^\mu d\tau - p_\mu \frac{c^2 d\tau \delta d\tau}{dx^\mu}\right) = \\
\int_{\tau_1}^{\tau_2} \left(F_\mu \delta x^\mu d\tau - mc^2 \delta d\tau\right) = 0,
\end{aligned}$$

where after using per-partes, we have used the fact, that we are working in a fixed-end problem. Finally, we have

$$\boxed{\int_{\tau_1}^{\tau_2} \left(F_\mu \delta x^\mu d\tau - mc^2 \delta d\tau\right) = 0}. \tag{1.17}$$

If we consider Hamilton's principle,

$$\delta S = \delta \int \mathcal{L} d\tau = \int \delta\mathcal{L} d\tau + \mathcal{L}\delta d\tau, \tag{1.18}$$

we can see, that for a free particle, with $F_\mu = 0$, we have

$$\mathcal{L}_{free,\tau} = -mc^2, \tag{1.19}$$

if we consider a proper time parametrization. However for a free particle with no external scalar field $\phi = 0$, we know that this m should be the usual rest mass. We see that if we take m to be the one defined in (1.4), then this holds. This fact presented here is non-trivial, since this is emanating from the definition of the four-momentum of the particle ($p^\mu = mu^\mu$, instead of $p^\mu = m_0 u^\mu$), however this choice will be clarified and justified later in the text.
In the equation (1.18) we can also substitute any other parametrization ζ. For the proper time, the following relation holds

$$d\tau = \frac{1}{c}\sqrt{-\eta_{\mu\nu} dx^\mu dx^\nu}, \tag{1.20}$$

and from it we can get a slightly modified version of the d'Alembert principle

$$\int \left(F_\mu \delta x^\mu d\tau - mc^2 \delta d\tau\right) = \delta \int \mathcal{L} d\tau = \delta \int \mathcal{L} \frac{d\tau}{d\zeta} d\zeta = \frac{1}{c}\delta \int \mathcal{L} \sqrt{-\eta_{\mu\nu} \frac{\mathrm{d}x^\mu}{\mathrm{d}\zeta} \frac{\mathrm{d}x^\nu}{\mathrm{d}\zeta}} d\zeta. \tag{1.21}$$

If we consider the free particle once again, we have

$$-\frac{1}{c}\delta\int mc^2\sqrt{-\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\zeta}\frac{\mathrm{d}x^\nu}{\mathrm{d}\zeta}}d\zeta=-\delta\int mc\sqrt{-\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\zeta}\frac{\mathrm{d}x^\nu}{\mathrm{d}\zeta}}d\zeta, \tag{1.22}$$

we can see, that the Lagrangian of a free particle with a different parametrization is

$$\mathcal{L}_{free,\zeta}=-mc\sqrt{-\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\zeta}\frac{\mathrm{d}x^\nu}{\mathrm{d}\zeta}}. \tag{1.23}$$

Where, again, this relation simplifies to (1.19) due to the normalization of four-velocity, if we take $\zeta=\tau$.
Now we can focus on the stated problem, where the particle is affected by the scalar field.
We start from the equation (1.17), where we substitute our definition of four-force from (1.2)

$$\int_{\tau_1}^{\tau_2}\left(F_\mu\delta x^\mu d\tau-mc^2\delta d\tau\right)=\int_{\tau_1}^{\tau_2}\left(-q\phi_{,\mu}\,\delta x^\mu d\tau-mc^2\delta d\tau\right)=$$
$$\int_{\tau_1}^{\tau_2}\left(-q\delta\phi d\tau-mc^2\delta d\tau\right)=\int_{\tau_1}^{\tau_2}\delta\left((-q\phi-mc^2)d\tau\right)+q\phi\delta d\tau+c^2d\tau\delta m.$$

Now we only need to "make a variation" out of the last two terms.

1.1.4 Action from the Lagrangian

Since m is dependent on the scalar field, the variation δm behaves as

$$\delta m=\frac{\mathrm{d}m}{\mathrm{d}\phi}\delta\phi. \tag{1.24}$$

Now we can continue with just the two last terms

$$\int_{\tau_1}^{\tau_2}q\phi\delta d\tau+c^2d\tau\delta m=\int_{\tau_1}^{\tau_2}q\phi\delta d\tau+c^2d\tau\frac{\mathrm{d}m}{\mathrm{d}\phi}\delta\phi=$$
$$\int_{\tau_1}^{\tau_2}q\phi\delta d\tau+qd\tau\delta\phi+\frac{\mathrm{d}k}{\mathrm{d}\phi}d\tau\delta\phi=q\int_{\tau_1}^{\tau_2}\delta(\phi d\tau)+\int_{\tau_1}^{\tau_2}\delta k d\tau.$$

Where in the second and third equality, the relations (1.24) and (1.3) were used respectively.
Thus, we get to the final integral (since $\delta k=0$)

$$\int_{\tau_1}^{\tau_2}\delta\left((-q\phi-mc^2+q\phi)d\tau\right)=\int_{\tau_1}^{\tau_2}\delta\left(-mc^2d\tau\right), \tag{1.25}$$

which would seem as a Lagrangian of a free particle, but we need to recall our definition of m. If we substitute it in from (1.4), we get

$$-c^2\int_{\tau_1}^{\tau_2}\delta\left((q\phi+m_0)d\tau\right)=0 \tag{1.26}$$

In total

$$\boxed{\mathcal{L}=-c^2(q\phi+m_0)}, \tag{1.27}$$

which is the same Lagrangian as in [16] .
If we transform τ into a general parametrization, we would get

$$\mathcal{L} = -c(q\phi + m_0)\sqrt{-\eta_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\zeta}\frac{\mathrm{d}x^\nu}{\mathrm{d}\zeta}}. \tag{1.28}$$

Now we can use the Lagrangian we got and find some equations of motion from it. One way to do it is using the variation of the action. Similar to [18], we can write the defining equation of motion (1.29) using (-,+,+,+) notation as

$$S[x(s)] = -m_0 \int ds(1 + q'\phi(x(s))\sqrt{-\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}, \tag{1.29}$$

where we have put the usual rest mass m_0 in front of the integral, and defined a new quantity corresponding to the specific charge of our particle $q' = \frac{q}{m_0}$.
We have derived the Lagrangian (1.28) in flat spacetime. However this result can be generalized to a curved background in a straightforward manner. In the following section, we will simply change $\eta \to g$ and use a proper time parametrization, effectively using the same Lagrangian as in (1.27).

1.2 Proper time variation

Using natural units, we will start from the action given by the Lagrangian derived in the previous section, given by(1.27)

$$S = -m_0 \int d\tau(1 + q'\phi). \tag{1.30}$$

The goal is to use the principle of least action, so we need to variate it

$$\delta S = -m_0 \int \delta(d\tau)(1 + q'\phi) + d\tau\delta(1 + q'\phi). \tag{1.31}$$

We will divide this into two separate integrals, since it is going to be a slightly longer calculation

$$\Xi_1 = \int \delta(d\tau)(1 + q'\phi)$$
$$\Xi_2 = \int d\tau\delta(1 + q'\phi)$$

where we have omitted the factor $-m_0$ which is without loss of generality, since we consider it to be the standard invariant proper mass and we can divide by it in the final expression since $\delta S = 0$. Starting with Ξ_1 we have

$$\Xi_1 = \int \delta(d\tau)(1 + q'\phi) =$$
$$\int \delta(\sqrt{-g_{\mu\nu}dx^\mu dx^\nu})(1 + q'\phi) =$$
$$\int \frac{1}{2d\tau}(-g_{\mu\nu,\alpha}\delta(x^\alpha)dx^\mu dx^\nu - 2g_{\mu\nu}dx^\mu\delta(dx^\nu))(1 + q'\phi)\cdot\frac{d\tau}{d\tau} =$$
$$\int d\tau\frac{1}{2}\left(-g_{\mu\nu,\alpha}\delta(x^\alpha)u^\mu u^\nu - 2g_{\mu\nu}u^\mu\frac{\mathrm{d}\delta(dx^\nu)}{\mathrm{d}\tau}\right)(1 + q'\phi)$$

Now, using per-partes only for the second term,

$$
\begin{aligned}
&-\int d\tau \left(g_{\mu\nu}u^{\mu}\frac{\mathrm{d}\delta(dx^{\nu})}{\mathrm{d}\tau}\right)(1+q'\phi) = \\
&\underbrace{-\left[(g_{\mu\nu}u^{\mu}\delta(x^{\nu}))(1+q'\phi)\right]}_{\to 0} + \int d\tau\delta(x^{\nu})\frac{\mathrm{d}\left((g_{\mu\nu}u^{\mu})(1+q'\phi)\right)}{\mathrm{d}\tau} = \\
&\int d\tau\delta(x^{\nu})\left(g_{\mu\nu,\alpha}u^{\alpha}u^{\mu}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\right)(1+q'\phi)+ \\
&\int d\tau\delta(x^{\nu})(g_{\mu\nu}u^{\mu})(q'\phi_{,\alpha}u^{\alpha})
\end{aligned}
$$

Adding the first term back again, we have

$$
\begin{aligned}
&\int d\tau\left(-\frac{1}{2}g_{\mu\nu,\alpha}\delta(x^{\alpha})u^{\mu}u^{\nu}\right)(1+q'\phi)+ \\
&\int d\tau\delta(x^{\nu})\left(g_{\mu\nu,\alpha}u^{\alpha}u^{\mu}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\right)(1+q'\phi)+ \\
&\int d\tau\delta(x^{\nu})(g_{\mu\nu}u^{\mu})(q'\phi_{,\alpha}u^{\alpha})) = \\
&\int d\tau\left(-\frac{1}{2}g_{\mu\nu,\alpha}\delta(x^{\alpha})u^{\mu}u^{\nu}+\delta(x^{\nu})\left(g_{\mu\nu,\alpha}u^{\alpha}u^{\mu}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\right)\right)(1+q'\phi)+ \\
&\int d\tau\delta(x^{\nu})(g_{\mu\nu}u^{\mu})(q'\phi_{,\alpha}u^{\alpha}))
\end{aligned}
$$

Now we can focus on the first integral here. The second one we will add later.

$$
\begin{aligned}
&\int d\tau\left(-\frac{1}{2}g_{\mu\nu,\alpha}\delta(x^{\alpha})u^{\mu}u^{\nu}+\delta x^{\nu}(g_{\mu\nu,\alpha}u^{\alpha}u^{\mu}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau})\right)(1+q'\phi) = \\
&\int d\tau\left((-\frac{1}{2}g_{\mu\nu,\alpha}+g_{\alpha\mu,\nu})u^{\mu}u^{\nu}\delta x^{\alpha}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\right)(1+q'\phi)
\end{aligned}
$$

Where we have renamed the indices in the second term as $\alpha \leftrightarrow \nu$ and used the fact that the metric is symmetric by definition. We can then symmetrize its derivative in the second term, since we are summing over a symmetric term. Rewriting this, we get to

$$
\begin{aligned}
&\int d\tau\left((-\frac{1}{2}g_{\mu\nu,\alpha}+g_{\alpha\mu,\nu})u^{\mu}u^{\nu}\delta x^{\alpha}+g_{\mu\nu}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\delta x^{\nu}\right)(1+q'\phi) = \\
&\int d\tau\left((-\frac{1}{2}g_{\mu\nu,\alpha}+\frac{1}{2}g_{\alpha\mu,\nu}+\frac{1}{2}g_{\alpha\nu,\mu})u^{\mu}u^{\nu}\delta x^{\alpha}+g_{\mu\alpha}\delta x^{\alpha}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}\right)(1+q'\phi) = \\
&\int d\tau\left(g_{\mu\alpha}\frac{\mathrm{d}u^{\mu}}{\mathrm{d}\tau}+\Gamma_{\alpha\mu\nu}u^{\mu}u^{\nu}\right)\delta x^{\alpha}(1+q'\phi) = \\
&\int d\tau g_{\beta\alpha}\left(\frac{\mathrm{d}u^{\beta}}{\mathrm{d}\tau}+\Gamma^{\beta}{}_{\mu\nu}u^{\mu}u^{\nu}\right)\delta x^{\alpha}(1+q'\phi) = \\
&\int g_{\beta\alpha}\left(\frac{Du^{\beta}}{d\tau}\right)\delta x^{\alpha}(1+q'\phi) = \\
&\int u_{\alpha;\mu}u^{\mu}\delta x^{\alpha}(1+q'\phi)
\end{aligned}
$$

And for the second variation we have

$$\Xi_2 = \int d\tau \delta(1 + q'\phi) = \int d\tau q' \phi_{,\alpha} \delta x^\alpha$$

So, in total we have

$$\int u_{\alpha;\mu} u^\mu \delta x^\alpha (1 + q'\phi) d\tau + \int \delta(x^\nu)(g_{\mu\nu} u^\mu)(q' \phi_{,\alpha} u^\alpha)) d\tau + \int d\tau q' \phi_{,\alpha} \delta x^\alpha =$$
$$\int u_{\alpha;\mu} u^\mu \delta x^\alpha (1 + q'\phi) d\tau + \int (\delta(x^\nu)(g_{\mu\nu} u^\mu)(q' \phi_{,\alpha} u^\alpha) + q' \phi_{,\alpha} \delta x^\alpha) d\tau$$

Now we know that should be equal to zero. Since it must hold for all variations, the integrand must be zero as well. We have

$$\begin{aligned}
&u_{\alpha;\mu} u^\mu \delta x^\alpha (1 + q'\phi) + (\delta(x^\nu)(g_{\mu\nu} u^\mu)(q' \phi_{,\alpha} u^\alpha) + q' \phi_{,\alpha} \delta x^\alpha) = 0 \\
&u_{\alpha;\mu} u^\mu \delta x^\alpha (1 + q'\phi) = -(\delta(x^\nu)(g_{\mu\nu} u^\mu)(q' \phi_{,\alpha} u^\alpha) + q' \phi_{,\alpha} \delta x^\alpha) \\
&u_{\alpha;\mu} u^\mu (1 + q'\phi) \delta x^\alpha = -(g_{\mu\alpha} u^\mu)(q' \phi_{,\iota} u^\iota) \delta x^\alpha - q' \phi_{,\alpha} \delta x^\alpha \\
&u_{\alpha;\mu} u^\mu (1 + q'\phi) g^{\alpha\nu} = -(g_{\mu\alpha} u^\mu)(q' \phi_{,\iota} u^\iota) g^{\alpha\nu} - q' \phi_{,\alpha} g^{\alpha\nu} \\
&u^\nu{}_{;\mu} u^\mu (1 + q'\phi) = -u^\nu (q' \phi_{,\iota} u^\iota) - q' \phi_{,\alpha} g^{\alpha\nu} \\
&u^\nu{}_{;\mu} u^\mu = -\frac{q'}{(1 + q'\phi)} (u^\nu u^\alpha + g^{\alpha\nu}) \phi_{,\alpha} \\
&u^\nu{}_{;\mu} u^\mu = -\frac{q'}{(1 + q'\phi)} (u^\nu u^\mu + g^{\mu\nu}) \phi_{,\mu}
\end{aligned}$$

Thus we have

$$\boxed{u^\nu{}_{;\mu} u^\mu = -\frac{q'}{(1 + q'\phi)} (u^\nu u^\mu + g^{\mu\nu}) \phi_{,\mu}}. \tag{1.32}$$

Let us now comment on its form.
The whole term on the left hand side is the absolute derivative of the four-velocity. The prefactor on the right hand side corresponds to a scaled dynamical mass, which we will discuss further on.
A term similar to $u^\nu u^\mu + g^{\mu\nu}$ is discussed in [16], let us show its main properties here.
If we multiply the term by $g_{\nu\iota}$, we get

$$(u^\nu u^\mu + g^{\mu\nu}) g_{\nu\iota} = u^\mu u_\iota + \delta^\mu_\iota,$$

since the term $g_{\nu\iota}$ is the inverse of $g^{\mu\nu}$.
We obtained the orthogonal projection to u^μ defined as $\Delta^\mu_\nu = \delta^\mu_\nu + u^\nu u_\nu$. Here we must state, that in this form, we work with our sign convention, whereas in [16], the second term has a minus sign in front of it.
Let us verify, that this is in fact a projector. The property of idempotence states $\Delta^\mu_\nu \Delta^\nu_\iota = \Delta^\mu_\iota$. Filling in the terms

$$(\delta^\mu_\nu + u^\mu u_\nu)(\delta^\nu_\iota + u^\nu u_\iota) = \delta^\mu_\iota + 2u^\mu u_\iota + u^\mu u^\nu u_\iota u_\nu = \delta^\mu_\iota + u^\mu u_\iota = \Delta^\mu_\iota,$$

we can see that this holds.
And the last factor in (1.32) is simply the gradient of the scalar field. Thus, the orthogonal projection acts on the gradient of the field. In other words, the produced force is non-trivial only in directions transversal to four-velocity.

1.2.1 Using dynamic mass

If we follow the same steps as in [16], we can define a new quantity (the same as our m previously), named dynamic mass

$$\mathcal{M} = m_0 + q\phi, \tag{1.33}$$

this represents the mass as a scalar, which is dependent on the coordinates.
We can now rewrite the Lagrangian, and subsequently the action to the form

$$S = -\int \mathcal{M} d\tau, \tag{1.34}$$

where now we have to integrate over the scalar mass $\mathcal{M}$. We are using calligraphic notation to differentiate it from a mass parameter appearing in the following metrics which will be denoted by the capital M. We variate the action to get

$$\delta S = -\int \mathcal{M}\delta d\tau + \delta\mathcal{M} d\tau, \tag{1.35}$$

but we have already solved a similar expression, since the first term is a rescaled integral Ξ_1 and the second term is the rescaled integral Ξ_2, both by a factor of m_0. In detail, for the first term, we again perform per-partes integration, use the fact that we are considering a problem with fixed ends and the whole first term becomes

$$\int u_{\alpha;\mu} u^\mu \delta x^\alpha \mathcal{M} d\tau + \int \delta x^\nu g_{\mu\nu} u^\mu \mathcal{M}_{,\alpha} u^\alpha d\tau. \tag{1.36}$$

And for the second term we easily get

$$\int \mathcal{M}_{,\mu} \delta x^\mu d\tau. \tag{1.37}$$

Combining these two, we get

$$\int u_{\alpha;\mu} u^\mu \delta x^\alpha \mathcal{M} d\tau + \int \delta x^\nu g_{\mu\nu} u^\mu \mathcal{M}_{,\alpha} u^\alpha d\tau + \int \mathcal{M}_{,\mu} \delta x^\mu d\tau = 0$$
$$u_{\alpha;\mu} u^\mu \mathcal{M} + g_{\mu\alpha} u^\mu \mathcal{M}_{,\iota} u^\iota + \mathcal{M}_{,\alpha} = 0$$

So we get the equation of motion

$$\boxed{\frac{D(\mathcal{M}u_\alpha)}{d\tau} = -\mathcal{M}_{,\alpha}}. \tag{1.38}$$

Where we have used the fact that $\mathcal{M}$ is a scalar and lowered the index using the metric.
We can use the same notation as in [16], the generalized momentum

$$P_\mu = \mathcal{M} u_\mu,$$

and substitute it into (1.38) to get

$$\frac{D}{d\tau}(P_\mu) = -\mathcal{M}_{,\mu}, \tag{1.39}$$

which has a clear form of an equation of motion.
The next step is defining the variable f_μ with the physical meaning of the mass density of the generalized force. So, for it, we can see

$$f_\mu = \frac{1}{\mathcal{M}}\dot{P}_\mu = \frac{1}{\mathcal{M}}(\dot{\mathcal{M}}u_\mu + \mathcal{M}a_\mu). \tag{1.40}$$

The motivation for defining this variable originates from the definition of force, using which we can reduce the order of the equation. From the view of classical mechanics, we have

$$\ddot{\vec{x}} = \frac{\vec{F}}{m} = \frac{\dot{\vec{p}}}{m}. \tag{1.41}$$

Now we want to have only the projection of the four-acceleration orthogonal to the four-velocity, which we get using the afore-defined projection operator Δ^μ_ν as

$$a^\mu = \ddot{x}^\mu = \Delta^\mu_\nu f^\nu. \tag{1.42}$$

We can see from the equation (1.40) that the equation (1.42) holds. The projection space of the operator is verified very easily, as the calculation uses only the normalization of four-velocity and the orthogonality of four-acceleration to four-velocity.

2 FJNW spacetime

2.1 The original solution

The Fisher-Janis-Newman-Winicour (or simply FJNW) spacetime has been introduced in its most familiar form in 1968 in a paper called Reality of the Schwarzschild singularity[8].
We will present a summary of the most important aspects (important to our field of study) of their findings. We will treat (r, θ, φ) as the usual spherical coordinates, thus $r \in [0, +\infty), \theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.
This spacetime is a spherically symmetric, static non-vacuum solution to the Einstein-scalar field equations. It has been intended to be an extended version of the Schwarzschild solution to further study singular event horizons. The solution is characterized by the following line element[8]

$$ds^2 = -\left(\frac{2R - r_0(\mu-1)}{2R + r_0(\mu+1)}\right)^{\frac{1}{\mu}} dt^2 + \left(\frac{2R + r_0(\mu+1)}{2R - r_0(\mu-1)}\right)^{\frac{1}{\mu}} dR^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2), \tag{2.1}$$

where R is a modified radial coordinate, $r_0 = 2M$ and μ is a parameter characteristic for the solution. And r is a function of R that is presented below. For the scalar field we have

$$\phi = \frac{A}{\mu} \ln \left|\frac{2R - r_0(\mu-1)}{2R + r_0(\mu+1)}\right|, \tag{2.2}$$

where A is a parameter characterizing the solution. Furthermore we have

$$r^2 = \frac{1}{4}[2R + r_0(\mu+1)]^{1+\frac{1}{\mu}}[2R - r_0(\mu-1)]^{1-\frac{1}{\mu}} \tag{2.3}$$

and

$$\mu = \sqrt{1 + 4\kappa\frac{A^2}{r_0^2}} \geq 1, \tag{2.4}$$

where κ is the coupling constant.
Now, since r was the original radial coordinate, and $r \in [0, +\infty)$, this corresponds to $R \in [\frac{1}{2}r_0(\mu-1), +\infty)$. There is a singularity at $R = \frac{1}{2}r_0(\mu-1)$, since the scalar potential, the Ricci tensor and the Ricci scalar all diverge[8].
The authors then discuss the important limit as $\mu \to 1$ (or $\kappa \to 0$). So we proceed similarly, checking the correspondence of R on r in this limit. The important term is the last one in (2.3)

$$J := [2R - r_0(\mu-1)]^{1-\frac{1}{\mu}}. \tag{2.5}$$

For $R = \frac{1}{2}r_0(\mu-1)$, this limit is undefined, since it behaves like $\lim_{\mu\to 1} 0^{1-\frac{1}{\mu}} = 0^0$. Since the limit is undefined, we can set it equal to some value and infer the results. So, if we impose the condition that (2.5) ($R \to 0$) be 1, then this corresponds to the Schwarzschild metric. Now $R = 0 \leftrightarrow r = 2M$. This can easily be seen from (2.3) if we take the limit $\mu \to 1$, then

$$r^2 = \frac{1}{4}(2R + 4M)^2 \to r = R + 2M. \tag{2.6}$$

A different condition would dictate that we take (2.5) to be 0 in the limit. For this limit $R = 0 \leftrightarrow r = 0$. But still for $R > 0$, the previous solution holds, thus $r = R + 2M$. So outside the 'radius' $r = 2M$, the solution is Schwarzschild, but inside, the space collapses to a single point.
We have to note, that this choice of the limit is purely arbitrary. If we were to take any positive value, we would just rescale the correspondence $r = R + 2M$. Taking a negative value would lead to non-physical results for both coordinates being real.

2.2 Reparametrized solution

We can also reparametrize this solution into a simpler form using different variables as listed below.
The original line element (2.1) can be transformed into[18]

$$ds^2 = -\left(1 - \frac{2M}{nr}\right)^n dt^2 + \left(1 - \frac{2M}{nr}\right)^{-n} dr^2 + \left(1 - \frac{2M}{nr}\right)^{1-n} r^2(d\theta^2 + \sin^2\theta d\varphi^2) \tag{2.7}$$

where M is the physical mass, n is the parameter characterizing the influence of scalar field, r is a modified radial coordinate and θ, φ correspond to the usual angles of spherical coordinates. It is immediately seen from this form that the solution reduces to the exterior Schwarzschild case for $n = 1$. It is important to state that $r \in [\frac{2M}{n}, +\infty)$, which we will comment on later.
This spacetime has a quasi-naked singularity, since at $r = 2M/n$, the term in front of the time component goes to zero, indicating a horizon, but this corresponds to a single point in spacetime where curvature scalars diverge.
Our focus will be mostly turned to $n \geq 0.5$, since below this value, the metric behaves very differently and would require a unique approach altogether. This complies with most treatments in other literature[18, 19] .
We will adopt the same notation as in [18], that is $f(r) = (1 - \frac{2M}{nr})$. We thus have

$$ds^2 = -f^n dt^2 + f^{-n} dr^2 + f^{1-n} r^2(d\theta^2 + \sin^2\theta d\varphi^2). \tag{2.8}$$

And the scalar field is given by

$$\phi(r) = \frac{\sqrt{1-n^2}}{2} \ln\left(1 - \frac{2M}{nr}\right) = \frac{\sqrt{1-n^2}}{2} \ln f. \tag{2.9}$$

We will need to use the metric with raised indices, so, since the metric $g_{\mu\nu}$ is diagonal in this solution, the inversion is straightforward. We get to

$$g_{\mu\nu} = \begin{pmatrix} -f^n & 0 & 0 & 0 \\ 0 & f^{-n} & 0 & 0 \\ 0 & 0 & r^2 f^{1-n} & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta f^{1-n} \end{pmatrix}. \tag{2.10}$$

And taking the inverse of this matrix, we have

$$g^{\mu\nu} = \begin{pmatrix} -f^{-n} & 0 & 0 & 0 \\ 0 & f^n & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} f^{n-1} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2\theta} f^{n-1} \end{pmatrix}. \tag{2.11}$$

Limit in the reparametrized solution

The FJNW spacetime is usually referred to as a spacetime with a naked singularity. But if we take $n = 1$ in (2.7), then this completely reduces to the exterior Schwarzschild solution. We will now explore for which n does the naked singularity appear.
To do that we need to find out how the parameter n is defined. If we compare the line elements from the original paper on FJNW spacetime[8] with the line element (2.7),then parameter $n = \frac{M}{\sqrt{M^2+q^2}}$ can be introduced[10]. Therefore, $n \in (0, 1]$.
After reparametrizing the metric, we will denote the radial coordinate in the solution as r and the areal radial coordinate as r' in this section. In general, it can be shown that the relation $r = r' + \sqrt{M^2 + q^2} + M$ holds[10].
If we want to retrieve the limit mentioned in the original paper, we need to see what μ corresponds to in these parameters. For μ we have[10] $\mu = \sqrt{1 + \frac{q^2}{M^2}}$. So $\mu \to 1$ means that $\frac{q}{M} \to 0$. Thus we have also $n \to 1$.
In this limit, the radial coordinate at the point $r = 2M$ corresponds to the spherical coordinate, with the following relation $r = r' + 2M$, so at this point $r' = 0$.
Now if we take the limit $n \to 0$, and set $r = 2M/n$, then we would have the same ambiguity in our limit as before. So we still have to take the limit to be 0 or 1 for $\mu \to 1$ in the original line element (2.1) and $n \to 0$ in the reparametrized solution (2.7). So the limit $\mu \to 1$ may be retrieved by $n \to 0$.

Now let us impose the FJNW metric inside our equation (1.32).

$$u^{\nu}{}_{;\mu}u^{\mu} = -\frac{q'}{(1+q'\phi)}(u^{\nu}u^{\mu} + g^{\mu\nu})\phi_{,\mu} \tag{2.12}$$

$$u^{\nu}{}_{;\mu}u^{\mu} = -\frac{q'}{(1+q'\frac{\sqrt{1-n^2}}{2}\ln f)}(u^{\nu}u^{\mu} + g^{\mu\nu})\left(\frac{\sqrt{1-n^2}}{2}\ln f\right)_{,\mu} \tag{2.13}$$

Now we will fix ν to a given index and investigate one of the four given equations. We adopt the convention $0, 1, 2, 3 = t, r, \theta, \varphi$ correspondingly.
Focusing on the gradient of the potential, we have in general

$$\phi_{,r} = \left(\frac{\sqrt{1-n^2}}{2}\ln\left(1 - \frac{2M}{nr}\right)\right)_{,r} = \frac{\sqrt{1-n^2}}{2}\frac{1}{\left(1-\frac{2M}{nr}\right)}\frac{2M}{nr^2} = \frac{M\sqrt{1-n^2}}{fnr^2}.$$

And for all the other non-radial components this is identically zero. However, in comparison with the results in [18], the authors present different results, specifically non-zero derivative for angular and time derivatives.
The covariant derivative is, by definition

$$u^{\nu}{}_{;\mu} = u^{\nu}{}_{,\mu} + \Gamma^{\nu}{}_{\mu\lambda}u^{\lambda}.$$

Since our metric is diagonal, the equations are greatly simplified, since many of the Christoffel symbols will vanish. Explicitly, the only non-zero ones are given

below

$$\Gamma^{\nu}{}_{\mu\mu} = -\frac{1}{2}g^{\nu\nu}(g_{\mu\mu})_{,\nu}$$
$$\Gamma^{\nu}{}_{\mu\nu} = (\ln\sqrt{|g_{\nu\nu}|})_{,\mu}$$
$$\Gamma^{\nu}{}_{\nu\nu} = (\ln\sqrt{|g_{\nu\nu}|})_{,\nu}$$

where $\nu \neq \mu$.
The fact that the others are identically zero is trivial. That the symbols satisfy these relations can be shown using their definition and contracting specific indices as we show below.

$$\Gamma^{\nu}{}_{\mu\mu} = \frac{1}{2}g^{\nu\alpha}(g_{\alpha\mu,\mu} + g_{\mu\alpha,\mu} - g_{\mu\mu,\alpha}) = \frac{1}{2}g^{\nu\nu}(g_{\nu\mu,\mu} + g_{\mu\nu,\mu} - g_{\mu\mu,\nu}) = -\frac{1}{2}g^{\nu\nu}g_{\mu\mu,\nu}$$

Where in the last line, we have used the fact that $\mu \neq \nu$.

$$\Gamma^{\nu}{}_{\mu\nu} = \frac{1}{2}g^{\nu\alpha}(g_{\alpha\mu,\nu} + g_{\nu\alpha,\mu} - g_{\mu\nu,\alpha}) = \frac{1}{2}g^{\nu\nu}(g_{\nu\mu,\nu} + g_{\nu\nu,\mu} - g_{\mu\nu,\nu}) =$$
$$\frac{1}{2}g^{\nu\nu}g_{\nu\nu,\mu} = \frac{1}{2}\frac{g_{\nu\nu,\mu}}{g_{\nu\nu}} = \frac{1}{2}(\ln|g_{\nu\nu}|)_{,\mu} = (\ln\sqrt{|g_{\nu\nu}|})_{,\mu}$$

Where we have also used the fact that the term $g_{\nu\nu} = \frac{1}{g^{\nu\nu}}$.

$$\Gamma^{\nu}{}_{\nu\nu} = \frac{1}{2}g^{\nu\mu}(g_{\nu\nu,\nu} + g_{\nu\mu,\nu} - g_{\nu\nu,\mu}) = g^{\nu\mu}g_{\nu\mu,\nu} - \frac{1}{2}g^{\nu\mu}g_{\nu\nu,\mu} = \frac{1}{2}g^{\nu\nu}g_{\nu\nu,\nu} =$$
$$\frac{1}{2}\frac{g_{\nu\nu,\nu}}{g_{\nu\nu}} = (\ln\sqrt{|g_{\nu\nu}|})_{,\nu}$$

But many more of these will vanish, since our metric is dependent only on the coordinates r and θ. We will always treat ν as a fixed index and then calculate the rest of the terms.
In general, all of the Christoffel symbols with the upper index set fixed will take the following form

$$\Gamma^{\nu}{}_{\mu\lambda} = \Gamma^{\nu}{}_{\mu\mu} + 2\Gamma^{\nu}{}_{\mu\nu} + \Gamma^{\nu}{}_{\nu\nu}, \tag{2.14}$$

where $\mu \neq \nu \neq \lambda$, and the factor 2 corresponds to the fact that they are symmetric in the last two indices. In this index notation, ν will be always treated as a fixed index. We will further explore some important aspects of this spacetime. We know the Ricci scalar can be computed from the gradient of the scalar field[18]

$$R_{\mu\nu} = 2\phi_{;\mu}\phi_{;\nu} = 2\phi_{,\mu}\phi_{,\nu}. \tag{2.15}$$

But we already know this derivative from earlier. We have

$$\phi_{,\mu} = \left(0, \frac{M\sqrt{1-n^2}}{fnr^2}, 0, 0\right), \tag{2.16}$$

so for the Ricci tensor, we get to

$$R_{\mu\nu} = \frac{2M(1-n^2)}{f^2n^2r^4}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.17}$$

And for the Ricci scalar, we have after contracting the indices

$$R = g^{\mu\nu}R_{\mu\nu} = g^{rr}R_{rr} = f^n\frac{M(1-n^2)}{f^2n^2r^4} = \frac{M(1-n^2)f^{n-2}}{n^2r^4},$$

so we clearly see that this diverges if $f \to 0$, which can happen for

$$r \to \frac{2M}{n},$$

since $n < 2$ as discussed earlier. So the singularity at this point is truly a curvature one. Note that the Ricci scalar does diverge also for $r \to 0$, but that is not a region inside this spacetime.

2.2.1 Calculating the covariant derivative

To be sure, we have checked a large part of the calculations in `Maple`. As we have previously stated, we will consider ν fixed in (1.32). Thus, in general, we have

$$u^{\nu}{}_{;\mu}u^{\mu} = -\frac{q'}{(1+q'\frac{\sqrt{1-n^2}}{2}\ln\left(1-\frac{2M}{nr}\right))}(u^{\mu}u^{\nu}+g^{\mu\nu})\left(\frac{\sqrt{1-n^2}}{2}\ln\left(1-\frac{2M}{nr}\right)\right)_{,\mu}.$$

The right hand side will be non-zero only if $\mu = r$. We will calculate the equations for different values of ν in the following text.
We will also introduce a factor $B(r)$ the same way as in [18], such that

$$B = \frac{2q'\sqrt{1-n^2}}{n(q'\sqrt{1-n^2}\ln f + 2)}. \tag{2.18}$$

And also, calculating the only non-zero derivative of the potential in advance, we recall that

$$\ln\left(1-\frac{2M}{nr}\right)_{,r} = \frac{2M}{nr^2f}.$$

So for all the equations, we can immediately write

$$u^{\nu}{}_{;\mu}u^{\mu} = -\frac{q'}{(1+q'\frac{\sqrt{1-n^2}}{2}\ln\left(1-\frac{2M}{nr}\right))}(u^{\mu}u^{\nu}+g^{\mu\nu})\left(\frac{\sqrt{1-n^2}}{2}\ln\left(1-\frac{2M}{nr}\right)\right)_{,\mu}$$
$$u^{\nu}{}_{;\mu}u^{\mu} = -Bn(u^{\mu}u^{\nu}+g^{\mu\nu})\frac{2M}{2nr^2f}$$
$$u^{\nu}{}_{;\mu}u^{\mu} = -B(u^{\mu}u^{\nu}+g^{\mu\nu})\frac{M}{r^2f}.$$

Time components

Let us begin with $\nu = t$. We have

$$u^{t}{}_{;\mu}u^{\mu} = -B(u^{r}u^{t}+g^{rt})\frac{M}{r^2f}$$
$$u^{t}{}_{;\mu}u^{\mu} = -B(u^{r}u^{t})\frac{M}{r^2f}.$$

Computation of the Christoffel symbols is shortened, since only the second term in (2.14) will play a role. We thus have

$$\Gamma^t{}_{rt} = (\ln\sqrt{|g_{tt}|})_{,r},$$

since the metric component g_{tt} is only dependent on r.
Differentiating, we get to (considering only $f \geq 0$ in this case)

$$\Gamma^t{}_{rt} = (\ln\sqrt{|g_{tt}|})_{,r} = \frac{1}{\sqrt{f^n}}\frac{1}{2\sqrt{f^n}}nf^{n-1}f_{,r} = \frac{n}{2f}f_{,r} = \frac{n}{2f}\frac{2M}{nr^2} = \frac{M}{fr^2}.$$

We can now substitute this into the original equation. Before doing that, we need to realize that Christoffel symbols are symmetric in the last two indices, so we don't have to write them all out. And finally, we can write the final expression for the time component

$$\begin{aligned}
&u^t{}_{,\mu}u^\mu + 2\Gamma^t{}_{rt}u^t u^r = -B(u^r u^t)\frac{M}{r^2 f}\\
&\frac{\mathrm{d}u^t}{\mathrm{d}x^\mu}\frac{\mathrm{d}\tau}{\mathrm{d}\tau}u^\mu + \frac{2M}{fr^2}u^t u^r = -B(u^r u^t)\frac{M}{r^2 f}\\
&\ddot{t} + \frac{2M}{fr^2}\dot{t}\dot{r}(1+\frac{1}{2}B) = 0.
\end{aligned}$$

Radial components

Now we set $\nu = r$. Again, we can write

$$u^r{}_{;\mu}u^\mu = -B(u^r u^r + g^{rr})\frac{M}{r^2 f}.$$

The middle term in (2.14) will vanish, since the term g_{rr} does not depend on any other coordinate than r.
For the rest of them, we have

$$\begin{aligned}
&\Gamma^r{}_{\mu\mu} = -\frac{1}{2}g^{rr}g_{\mu\mu,r}\\
&\Gamma^r{}_{rr} = (\ln\sqrt{|g_{rr}|})_{,r} = (\ln f^{-\frac{n}{2}})_{,r} = -\frac{1}{f^{-\frac{n}{2}}}\frac{n}{2}f^{-\frac{n}{2}-1}\frac{2M}{nr^2} = -\frac{M}{fr^2}.
\end{aligned}$$

Now we need to calculate the terms $g_{tt,r}$, $g_{\theta\theta,r}$, $g_{\varphi\varphi,r}$

$$\begin{aligned}
&g_{tt,r} = (-f^n)_{,r} = -nf^{n-1}\frac{2M}{nr^2}\\
&g_{\theta\theta,r} = (r^2 f^{1-n})_{,r} = 2rf^{1-n} + r^2(1-n)f^{-n}\frac{2M}{nr^2} = f^{-n}\left(2fr + (1-n)\frac{2M}{n}\right)\\
&g_{\varphi\varphi,r} = (r^2\sin^2(\theta)f^{1-n})_{,r} = \sin^2(\theta)g_{\theta\theta,r}.
\end{aligned}$$

In total we have

$$\begin{aligned}
&\Gamma^r{}_{\mu\mu}u^\mu u^\mu =\\
&-\frac{1}{2}f^n\left[-nf^{n-1}\frac{2M}{nr^2}u^t u^t + f^{-n}\left(2fr + (1-n)\frac{2M}{n}\right)(u^\theta u^\theta + \sin^2(\theta)u^\varphi u^\varphi)\right].
\end{aligned}$$

So our equation (1.32) for the radial component takes the form

$$
\begin{aligned}
&u^r{}_{,\mu}u^\mu + \Gamma^r{}_{\mu\mu}u^\mu u^\mu + \Gamma^r{}_{rr}u^r u^r = \\
&-\frac{q'}{(1+q'\frac{\sqrt{1-n^2}}{2}\ln f)}(u^r u^r + g^{rr})\left(\frac{\sqrt{1-n^2}}{2}\ln f\right)_{,r} \\
&\ddot{r} - \frac{1}{2}f^n\left[-nf^{n-1}\frac{2M}{nr^2}u^t u^t + f^{-n}\left(2fr + (1-n)\frac{2M}{n}\right)(u^\theta u^\theta + \sin^2(\theta)u^\varphi u^\varphi)\right] \\
&-\frac{M}{fr^2}u^r u^r = -\frac{q'}{(1+q'\frac{\sqrt{1-n^2}}{2}\ln f)}(\dot{r}\dot{r} + f^n)\frac{M\sqrt{1-n^2}}{fnr^2}.
\end{aligned}
$$

First we will work on simplifying the left hand side

$$
\begin{aligned}
&\ddot{r} - \frac{1}{2}f^n\left(-nf^{n-1}\frac{2M}{nr^2}u^t u^t\right) - \\
&\left(fr + (1-n)\frac{M}{n}\right)(u^\theta u^\theta + \sin^2(\theta)u^\varphi u^\varphi) - \frac{M}{fr^2}u^r u^r \\
&\ddot{r} - \frac{1}{2}\left[-f^{2n-1}\frac{2M}{r^2}u^t u^t + \left(2fr + (1-n)\frac{2M}{n}\right)(u^\theta u^\theta + \sin^2(\theta)u^\varphi u^\varphi)\right] - \frac{M}{fr^2}u^r u^r \\
&\ddot{r} + f^{2n-1}\frac{M}{r^2}\dot{t}^2 - \left(fr + (1-n)\frac{M}{n}\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) - \frac{M}{fr^2}\dot{r}^2 \\
&\ddot{r} + f^{2n-1}\frac{M}{r^2}\dot{t}^2 - \left(r - \frac{M}{n} - M\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) - \frac{M}{fr^2}\dot{r}^2.
\end{aligned}
$$

And for the right hand side

$$
-B(\dot{r}^2 + f^n)\frac{M}{fr^2}.
$$

Putting them back together, we get

$$
\begin{aligned}
&\ddot{r} + f^{2n-1}\frac{M}{r^2}\dot{t}^2 - \left(r - \frac{M}{n} - M\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) - \frac{M}{fr^2}\dot{r}^2 = \\
&-B(\dot{r}^2 + f^n)\frac{M}{fr^2} \\
&\ddot{r} + f^{2n-1}\frac{M}{r^2}\dot{t}^2 + B\frac{f^{n-1}M}{r^2} - \left(r - \frac{M+Mn}{n}\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) = \\
&-B(\dot{r}^2)\frac{M}{fr^2} + \frac{M}{fr^2}\dot{r}^2 \\
&\ddot{r} + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B) - \frac{M\dot{r}^2}{fr^2}(1-B) + \left(\frac{M(1+n)}{n} - r\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) = 0.
\end{aligned}
$$

Angular components

Now we can infer the angular components of our equation. We will start with θ. Again, we have

$$
\begin{aligned}
u^\theta{}_{;\mu}u^\mu &= -B(u^r u^\theta + g^{r\theta})\frac{M}{r^2 f} \\
u^\theta{}_{;\mu}u^\mu &= -B(u^r u^\theta)\frac{M}{r^2 f}.
\end{aligned}
$$

For the Christoffel symbols, we have only two non-zero (accounting for their symmetry in the last two indices)

$$\Gamma^{\theta}_{\ \varphi\varphi} = -\frac{1}{2}g^{\theta\theta}(g_{\varphi\varphi})_{,\theta} = -\frac{1}{2}\frac{1}{r^2}f^{n-1}r^2 2\sin(\theta)\cos(\theta)f^{1-n} = -\sin(\theta)\cos(\theta)$$

$$\Gamma^{\theta}_{\ r\theta} = \left(\ln\left(\sqrt{|g_{\theta\theta}|}\right)\right)_{,r} = \left(\ln\left(\sqrt{|r^2 f^{1-n}|}\right)\right)_{,r} =$$

$$\frac{1}{2}\frac{1}{r^2 f^{1-n}}(2rf^{1-n} + r^2(1-n)f^{-n}\frac{2M}{nr^2}) = \frac{1}{r} + \frac{M}{nfr^2}(1-n).$$

In total, the left hand side looks like

$$\ddot{\theta} - \sin(\theta)\cos(\theta)\dot{\varphi}^2 + 2(\frac{1}{r} + \frac{M}{nfr^2}(1-n))\,\dot{r}\dot{\theta}$$

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + 2(\frac{1}{r} + \frac{M}{nfr^2}(1-n))\,\dot{r}\dot{\theta}.$$

The full equation then is

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + 2\left(\frac{1}{r} + \frac{M}{nfr^2}(1-n)\right)\dot{r}\dot{\theta} = -B\dot{r}\dot{\theta}\frac{M}{r^2 f}$$

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + \frac{2M\dot{\theta}\dot{r}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{1}{n} + \frac{fr}{M}\right) = 0$$

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + \frac{2M\dot{\theta}\dot{r}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{1}{n} + \frac{r}{M} - \frac{2}{n}\right) = 0$$

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + \frac{2M\dot{\theta}\dot{r}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{r}{M} - \frac{1}{n}\right) = 0.$$

Finally, let us turn our attention on the last coordinate ϕ. Now we have

$$u^{\varphi}_{\ ;\mu}u^{\mu} = -B(u^{\varphi}u^{r} + g^{\varphi r})\frac{M}{r^2 f}$$

$$u^{\varphi}_{\ ;\mu}u^{\mu} = -B(u^{\varphi}u^{r})\frac{M}{r^2 f}.$$

Christoffel symbols are then

$$\Gamma^{\varphi}_{\ r\varphi} = \left(\ln\sqrt{|g_{\varphi\varphi}|}\right)_{,r} = \left(\ln\sqrt{|r^2\sin^2(\theta)f^{1-n}|}\right)_{,r} =$$

$$\frac{1}{2}\frac{1}{r^2\sin^2(\theta)f^{1-n}}(2r\sin^2(\theta)f^{1-n} + r^2\sin^2(\theta)(1-n)f^{-n}\frac{2M}{nr^2}) = \frac{1}{r} + (1-n)\frac{M}{nfr^2}$$

$$\Gamma^{\varphi}_{\ \theta\varphi} = \left(\ln\sqrt{|g_{\varphi\varphi}|}\right)_{,\theta} = \left(\ln\sqrt{|r^2\sin^2(\theta)f^{1-n}|}\right)_{,\theta} =$$

$$\frac{1}{2}\frac{1}{r^2\sin^2(\theta)f^{1-n}}(2r^2\sin(\theta)\cos(\theta)f^{1-n}) = \cot(\theta).$$

So, putting them together, we have

$$\ddot{\varphi} + 2\left(\frac{1}{r} + (1-n)\frac{M}{nfr^2}\right)\dot{r}\dot{\varphi} + 2\cot(\theta)\dot{\varphi}\dot{\theta} = -B(u^{\varphi}u^{r})\frac{M}{r^2 f}$$

$$\ddot{\varphi} + 2\cot(\theta)\dot{\varphi}\dot{\theta} + \frac{2M\dot{r}\dot{\varphi}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{1}{n} + \frac{fr}{M}\right) = 0$$

$$\ddot{\varphi} + 2\cot(\theta)\dot{\varphi}\dot{\theta} + \frac{2M\dot{r}\dot{\varphi}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{r}{M} - \frac{1}{n}\right) = 0.$$

Let us now write them all out for clarity

$$\ddot{t} + \frac{2M}{fr^2}\dot{t}\dot{r}\left(1 + \frac{1}{2}B\right) = 0 \tag{2.19}$$

$$\ddot{r} + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B) - \frac{M\dot{r}^2}{fr^2}(1 - B) + \left(\frac{M(1+n)}{n} - r\right)(\dot{\theta}^2 + \sin^2(\theta)\dot{\varphi}^2) = 0 \tag{2.20}$$

$$\ddot{\theta} - \frac{1}{2}\sin(2\theta)\dot{\varphi}^2 + \frac{2M\dot{\theta}\dot{r}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{r}{M} - \frac{1}{n}\right) = 0 \tag{2.21}$$

$$\ddot{\varphi} + 2\cot(\theta)\dot{\varphi}\dot{\theta} + \frac{2M\dot{r}\dot{\varphi}}{fr^2}\left(\frac{1}{2}B - 1 + \frac{r}{M} - \frac{1}{n}\right) = 0. \tag{2.22}$$

We can see, that if we did not have the minus sign in (1.32), and rescaled B by a factor of two, our equations would be the same as in[18].

2.2.2 Numerical solution of the FJNW equations of motion

Now without loss of generality, we can set the initial conditions as $\theta = \pi/2$ and $\dot{\theta} = 0$ at a time $t = 0$. We have freedom in setting up the initial spherical part of the coordinates, and can therefore orient the Cartesian axes orthogonal with respect to this plane $\theta = \pi/2$.
From the differential equation for the θ component(2.21), we see that now $\ddot{\theta} = 0$. So the motion is happening all in the ecliptic plane, since $\theta = 0$ is constant. The equations have now been simplified into the following form

$$\ddot{t} + \frac{2M}{fr^2}\dot{t}\dot{r}\left(1 + \frac{1}{2}B(r)\right) = 0 \tag{2.23}$$

$$\ddot{r} + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B(r)) - \frac{M\dot{r}^2}{fr^2}(1 - B(r)) + \left(\frac{M(1+n)}{n} - r\right)\dot{\varphi}^2 = 0 \tag{2.24}$$

$$\ddot{\theta} = 0 \tag{2.25}$$

$$\ddot{\varphi} + \frac{2M\dot{r}\dot{\varphi}}{fr^2}\left(\frac{1}{2}B(r) - 1 + \frac{r}{M} - \frac{1}{n}\right) = 0. \tag{2.26}$$

Here we have explicitly stated that $B = B(r)$. We see that these equation possess two cyclic coordinates, namely t and φ. This implies that we have two integrals of motion that correspond to the conservation of energy and the conservation of the φ-component of angular momentum respectively. Thus the initial condition for these coordinates is purely arbitrary - we can always translate them without changing the evolution of the system.
We have illustrated the steps also for one important aspect. The equations of motion we calculated differ from [18]. We obtained a different solution in(1.32), where we have a minus sign on the right hand side unlike the same equation in the afore mentioned paper. Additionally in the equations of motion (2.19, 2.20, 2.21, 2.22) the factor B differs by a factor of two, although its definition is identical (2.18).
If we set the angular component of the coordinates and four-velocity to be zero, $\varphi = 0$ and $\dot{\varphi} = 0$, at a time $t = 0$, it has an immediate effect on the second derivative $\ddot{\varphi} = 0$.

If we consider these conditions, the equations transform into a much easier form

$$\ddot{t} + \frac{2M}{fr^2}\dot{t}\dot{r}\left(1 + \frac{1}{2}B(r)\right) = 0 \tag{2.27}$$

$$\ddot{r} + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B(r)) - \frac{M\dot{r}^2}{fr^2}(1 - B(r)) = 0 \tag{2.28}$$

$$\ddot{\theta} = 0 \tag{2.29}$$

$$\ddot{\varphi} = 0, \tag{2.30}$$

this means, we need to analyse only two second order coupled differential equations. We can do another sanity check, if we set $n = 1$ and $q' = 0$, we should be able to retrieve Schwarzschild equations of motion. Doing so for the radial coordinate, we have $B = 0$, thus

$$\ddot{r} = -\frac{M}{r^2}f\dot{t}^2 + \frac{M\dot{r}^2}{fr^2} - (2M - r)\,\dot{\varphi}^2,$$

which is exactly Schwarzschild.
All these and the following equations have been solved numerically, using the function `solve_ivp` from Python's library Scipy. Codes which are non-trivial and used in this work are addressed in the attachment section.
However there is a numerical issue in the $B(r)$ term. We have checked the denominator of $B(r)$ using Sympy, and it becomes zero for $r \approx 7.45$ for our specific choice of initial values, that is $M = 1$, $q = 3$ and $n = 0.5$. The analytical form of the critical value r_C for this solution is

$$r_C = \frac{2M\exp\left\{\frac{2}{q'\sqrt{1-n^2}}\right\}}{n\left(\exp\left\{\frac{2}{q'\sqrt{1-n^2}}\right\} - 1\right)}. \tag{2.31}$$

This function corresponds to the term $1 + q'\phi = 0$. If we plot this function as a function of q', we see that the the critical radius r_C becomes negative for negative values of q'. The limit $q' \to 0^-$ is zero, but the limit $q' \to 0^+$ gives $r_C = 2M/n$.

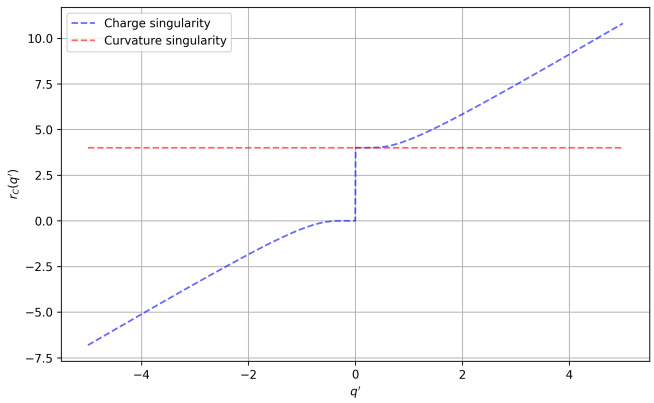


Figure 2.1 Critical radius r_C as a function of charge q'. Constants are set to $M = 1$, $n = 0.5$

The critical radius corresponds to the singularity at $r = 2M/n$ only for $q' = 0$. As $q' > 0$, r_C is larger than the point where our singularity is located, so we

cannot explore the motion there. On the other hand, if we take $q' < 0$, the critical value becomes negative, so the term does not blow up.
Near this critical radius, the right hand side of the equation (1.32) diverges, so the radial part of the covariant derivative diverges to negative infinity, since the derivative of the potential is positive. That means there is another horizon located at $r = r_C$, representing the **'charge' horizon**, or the charge singularity.
However from a purely physical aspect, this means that the factor $1 + q'\phi = 0$, which implies that the dynamic mass of the particle is zero. This is because for massive particles, we have in this sphere $m_0 + q\phi = 0$.
To illustrate the equations, we started by setting initial four-velocity in the t,r and φ components. From there we calculated the initial θ component using the normalization of four-velocity, which stays constant for the duration of the motion. That has been used as a sanity-check in the process of the numerical calculation, however it cannot be fulfilled to full precision. We have tried to bypass the problem by setting constants and initial velocities to lower values so as not to encounter problems in numerical approximation, but still, after many iterations, the normalization of four-velocity simply can not be fulfilled. Nevertheless it was possible for the initial iterations, which are crucial to our problem. To improve this situation, one would have to consider different formulation of the problem, potentially combined with more powerful numerical methods.
Here we illustrate the motion in Cartesian coordinates. The pink arrow represents the initial velocity, whose magnitude has been scaled for visibility. We should now recall that the curvature singularity at $r = 2M/n$ is just a single point, but we are representing the evolution of the whole system in Cartesian coordinates, so it looks as if though it is a ball of that radius. This means that the region inside the given ball does not exist in our coordinates. However this is not true for the charge singularity, which is a sphere of radius r_C (only in a given mode of $q' > 0$, otherwise it does not exist). We will also compare the motion to the Schwarzschild case, for which we just set the constants to $q' = 0$ and $n = 1$. In this case, the radius $r = 2M$ does not represent the curvature singularity, but it is in fact the event horizon. Thus it is not a point in spacetime, but a sphere inside which is the singularity at $r = 0$.

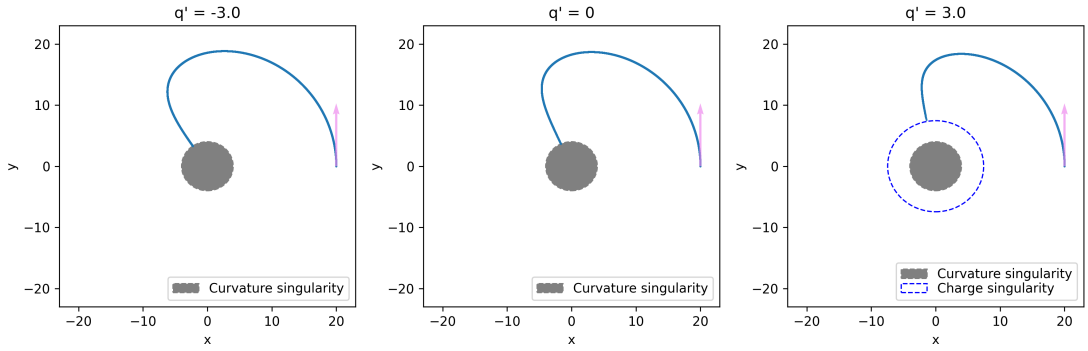


Figure 2.2 Geodesics in FJNW spacetime with $n = 0.5$, $M = 1$.

So we see that in the case of $q' = 3$ (an extremely large charge), there is a charge singularity enclosing the curvature one. The motion terminates in it and the equation for geodesic motion locally breaks down. In the next section, we will introduce the method of effective potential to further explore the motion near this

sphere.
We have also noticed that there exists a sphere of a varying radius which exists only for some specific combination of the initial parameters, inside which the radial acceleration is positive, i.e. out of the centre. We will explore this unusual behaviour further below. For FJNW the equations of motion are so complex (even if we were to set $\varphi = 0$ throughout the whole motion), that the analytical solution of this radius is very hard to find. We will try to study the system more systematically in its ultrastatic limit discussed below. It makes sense to think of this area as a ball, or rather in our case a spherical shell, whose innermost radius is a part of the particles trajectory where the acceleration changes sign. This is because we have spherical symmetry and can rotate the whole picture around the angle φ. Also keep in mind that we are presenting the radius only inside the area where the particle has travelled in the given evolution, which may not cover the whole space where the particle would be repelled away. Plotting for the same initial values as in Fig2.2, we can see the repulsive radius.

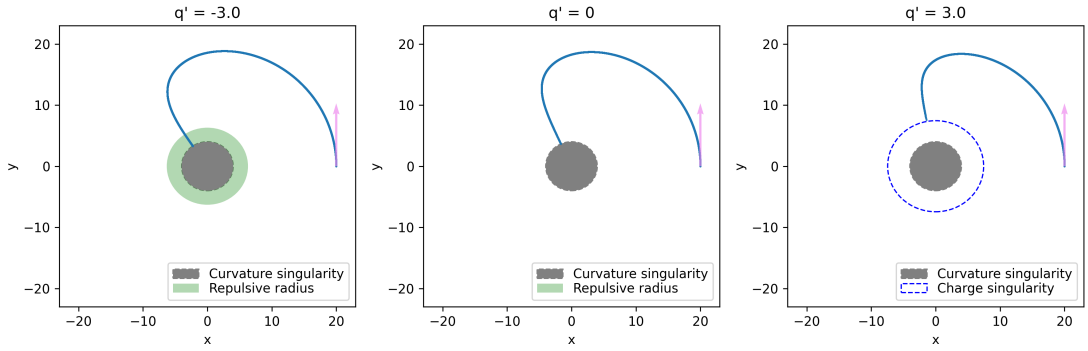


Figure 2.3 Geodesics in FJNW spacetime illustrating the repulsive radius with $n = 0.5$, $M = 1$.

The particle (or any object for that matter) carrying negative charge experiences a repulsive force near the centre.
Now, considering the same n and M, but for larger $\dot{\varphi}$, we get the following figure.

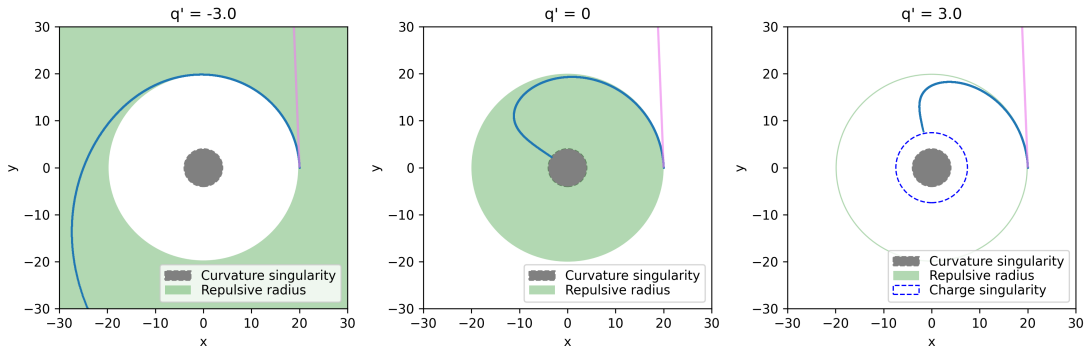


Figure 2.4 Geodesics in FJNW spacetime illustrating the repulsive radius with $n = 0.5$, $M = 1$ for large initial $\dot{\varphi}$.

We see that the negatively charged particle and the uncharged particle are both experiencing a resulting outward force. However for the uncharged particle, this force is not sufficient to repel it away form the curvature singularity, since the

initial $\dot{r}$ is negative and large enough in magnitude. In contrast, the negatively charged particle is repelled away, so the resulting force is larger in this case. The positively charged particle falls into the charge singularity, however it experienced the resulting repulsive force in the initial part of its motion due large $\dot{\varphi}$.
Considering lower charges, we have the following figure.

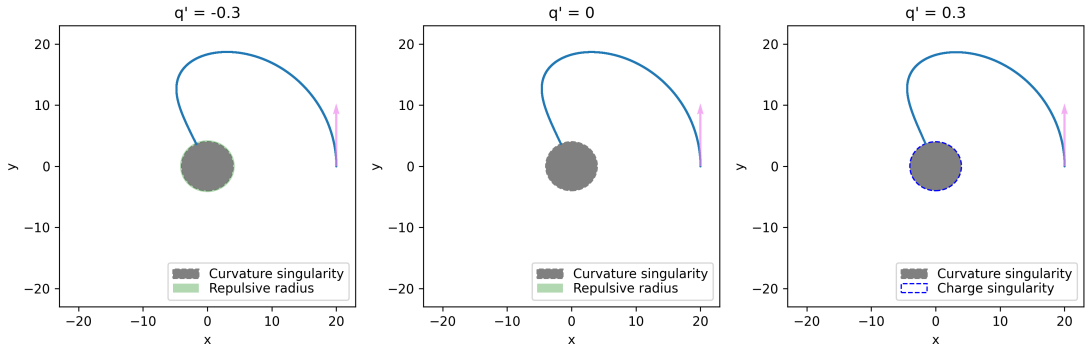


Figure 2.5 Geodesics in FJNW spacetime illustrating the repulsive radius with $n = 0.5$, $M = 1$ for lower charges.

As mentioned above, the charge horizon still encloses the curvature singularity for positive (albeit very small) charge. The particle possessing negative charge, moving near the centre is still repelled away, however the repulsive radius becomes smaller in size. This complies with Fig.2.1.
We can also compare the uncharged case with the usual Schwarzschild geodesics, where the positive second derivative corresponds to a centrifugal-like term. We get this special case by setting $q' = 0$ and $n = 1$. Plotting with the same values, except in the Schwarzschild spacetime, we see a different behaviour.

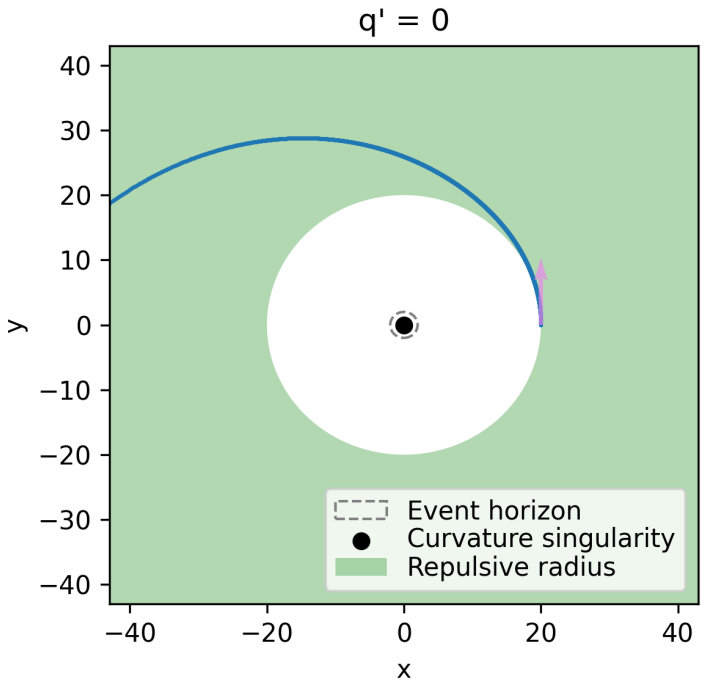


Figure 2.6 Geodesic in Schwarzschild spacetime illustrating the repulsive radius with $M = 1$.

So we see that throughout the geodesic motion, the angular term dominates and the particle does not reach the event horizon for the same initial conditions. Changing the initial radius to be inside the charge singularity, which has meaning only for the positive charge, nevertheless for the other cases this illustrates the motion well, we get the following figure.

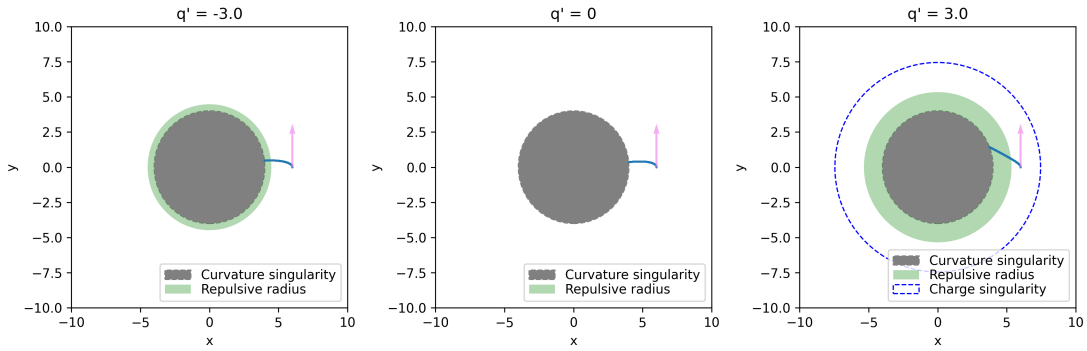


Figure 2.7 Geodesics in FJNW spacetime illustrating the repulsive radius inside the charge singularity with $n = 0.5$, $M = 1$.

For the negative charge and no charge nothing changes as expected. However, for the particle carrying positive charge, there appears a repulsive radius near the centre.
Now, once more, illustrating the Schwarzschild case for comparison.

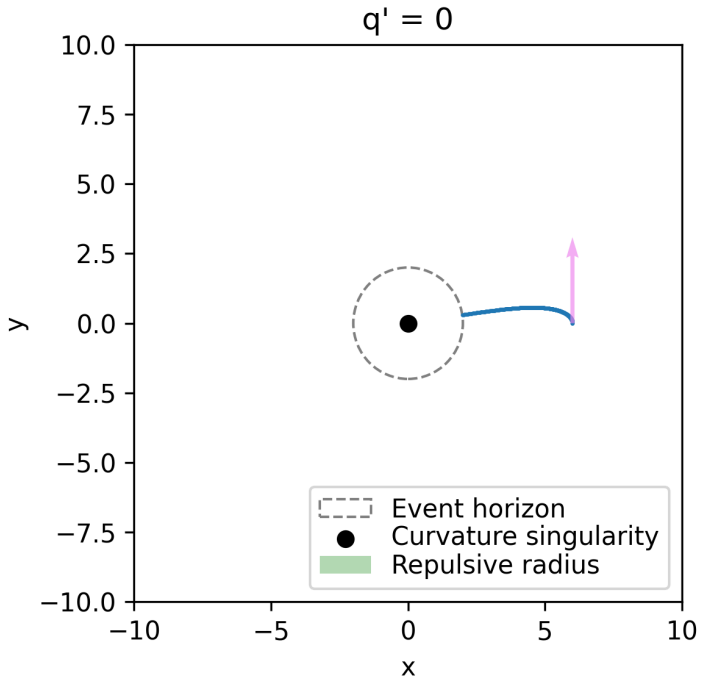


Figure 2.8 Geodesic in Schwarzschild spacetime illustrating the repulsive radius inside the original charge singularity with $M = 1$.

The particle then simply falls into the event horizon without being repelled. The motion does not visibly differ from the FJNW spacetime for an uncharged particle.

Since $n \in (0, 1]$, we can explore the motion for different values. However, as mentioned above, we will focus only on $n > 0.5$. Now, setting $n = 0.8$, we get the following figure.

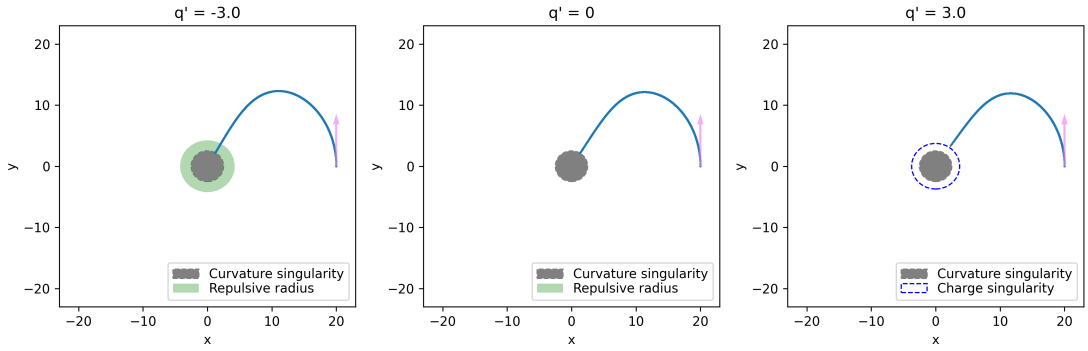


Figure 2.9 Geodesics in FJNW spacetime with $n = 0.8$, $M = 1$.

With n now set to 0.8, the charge singularity has a lower radius and the whole system has a higher attractive force.
When the particle starts its motion inside the charge horizon, we have the following figure.

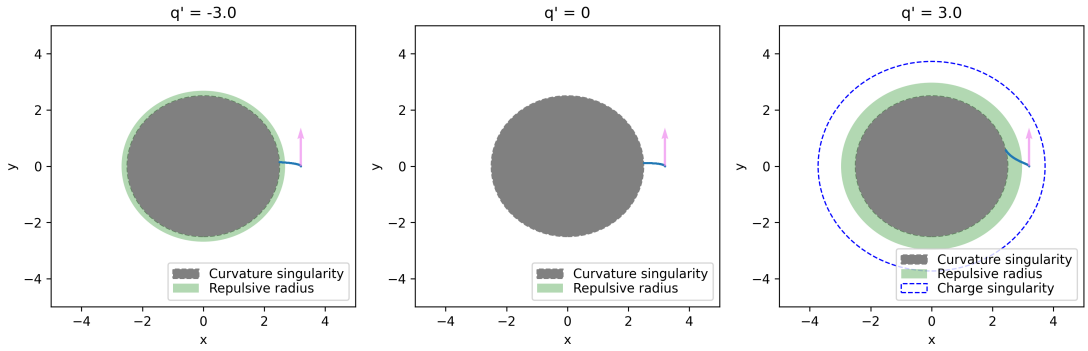


Figure 2.10 Geodesics in FJNW spacetime inside the charge horizon with $n = 0.8$, $M = 1$.

The particle carrying positive charge is still repelled near the curvature singularity, which also holds for the particle with negative charge, although the repulsive radius for the negatively charged particle is smaller.
In summary, the influence of the particle's charge is very limited and, with respect to the trajectories discussed above, effectively negligible.

2.2.3 Circular orbits in FJNW

We will now focus on a circular motion in FJNW spacetime. For that, the radial coordinate must not change $\dot{r} = 0$. Taking that into account, the equations

of motion would reduce to

$$\begin{aligned}
&\ddot{t} = 0 \\
&0 + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B(r)) + \left(\frac{M(1+n)}{n} - r\right)\dot{\varphi}^2 = 0 \\
&\ddot{\theta} = 0 \\
&\ddot{\varphi} = 0,
\end{aligned}$$

so we get just an equation for the time and angular components of four-velocity

$$\begin{aligned}
&0 + \frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B(r)) + \left(\frac{M(1+n)}{n} - r\right)\dot{\varphi}^2 = 0 \\
&\left(\frac{M(1+n)}{n} - r\right)\dot{\varphi}^2 = -\frac{f^{n-1}M}{r^2}(f^n\dot{t}^2 + B(r)).
\end{aligned}$$

But all of these are defined solely by the initial conditions, so we can calculate for example how to set $\dot{\varphi}$ to get circular orbits

$$\dot{\varphi} = \sqrt{-\frac{f^{n-1}M}{\left(\frac{M(1+n)}{n} - r\right)r^2}(f^n\dot{t}^2 + B(r))}, \tag{2.32}$$

since $B(r)$ is a constant term.
In practice, we set the initial velocity $\dot{t}$ and the radial distance r to some specific value and calculate $\dot{\varphi}$ from this expression. For $\dot{\theta}$ and $\dot{r}$ we use the normalization of four-velocity.
Doing this, we obtained circular orbits, although they have slightly different values of $\dot{\varphi}$ for $q' = -3, 0, 3$. Even though they are very similar for different values of q', the seventh digit after the decimal dot plays a role whether the motion terminates in infinity, the singularity, or as a circular orbit, suggesting unstable orbits. Below we illustrate the circular motion specifically calculated for $q' = -3$ and the quasi-circular motion for the same initial variables for $q' = 0, 3$.

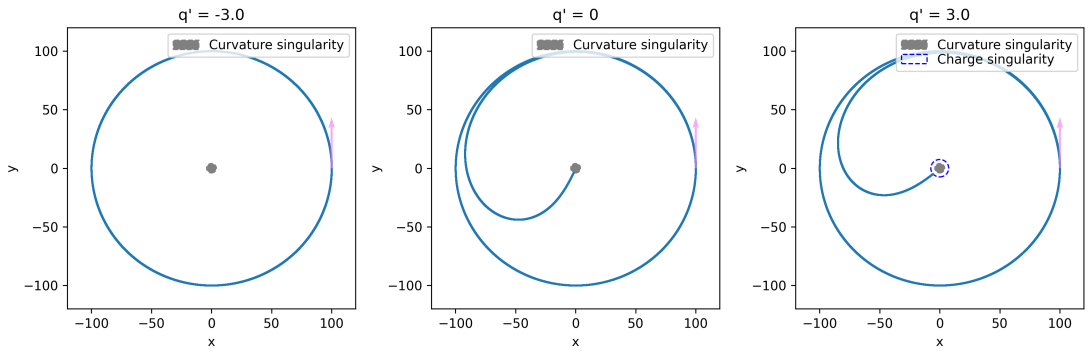


Figure 2.11 Circular motion in FJNW spacetime for $q' = -3$ with $n = 0.5$, $M = 1$.

Thus the negatively charged particle is acted on with a very slightly larger repulsive force than for the other two cases.
Next, calculating for the uncharged case, we get the following figure.

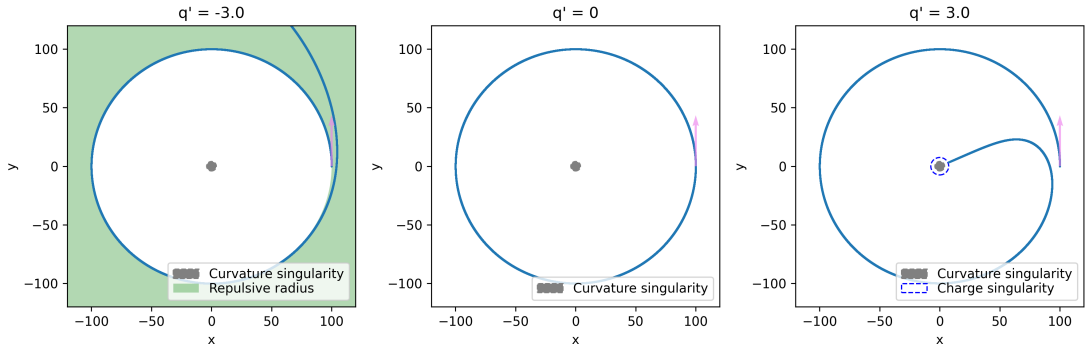


Figure 2.12 Circular motion in FJNW spacetime for $q' = 0$ with $n = 0.5$, $M = 1$.

The negatively charged particle for this radius is repelled away. However the positively charged particle still falls inside the charge horizon.
And for the particle with positive charge we have the following.

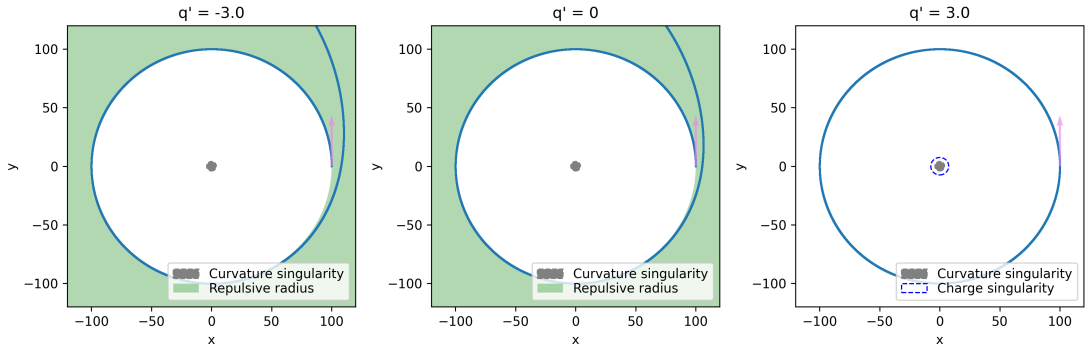


Figure 2.13 Circular motion in FJNW spacetime for $q' = 3$ with $n = 0.5$, $M = 1$.

From here we can finally infer, that **positive charge corresponds to an attractive force**. However as we can see from (2.32), the difference is very small, as the charge is manifested only the $B(r)$ term.

2.3 Ultrastatic limit of FJNW

A spacetime is called **ultrastatic** if we can write the infinitesimal spacetime interval as

$$ds^2 = -dt^2 + g_{ij}dx^i dx^j, \tag{2.33}$$

where the metric coefficients do not depend on t [20]. Ultrastatic spacetimes are not common in physics, but they are useful for simplifications of some calculations, such as in understanding the Casimir effect and the Hadamard condition.
The line element in FJNW spacetime (2.1) can be rewritten after explicitly writing out the function $r(R)$ as

$$ds^2 = -\left(\frac{2R - r_0(\mu-1)}{2R + r_0(\mu+1)}\right)^{\frac{1}{\mu}} dt^2 + \left(\frac{2R + r_0(\mu+1)}{2R - r_0(\mu-1)}\right)^{\frac{1}{\mu}} dR^2 +$$
$$\frac{1}{4}[2R + r_0(\mu+1)]^{1+\frac{1}{\mu}}[2R - r_0(\mu-1)]^{1-\frac{1}{\mu}}(d\theta^2 + \sin^2\theta d\varphi^2).$$

Now we will introduce a factor $f(R,\mu) = \left(\frac{2R - r_0(\mu-1)}{2R + r_0(\mu+1)}\right)$.
The line element now looks accordingly

$$ds^2 = -f(R,\mu)^{\frac{1}{\mu}} dt^2 + f(R,\mu)^{-\frac{1}{\mu}} dR^2 + \frac{1}{4}(2R + r_0(\mu+1))(2R - r_0(\mu-1)) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2).$$

To simplify this, we will introduce a rescaled factor $\tilde{r_0} = r_0/2$. This transforms the line element into

$$ds^2 = -f(R,\mu)^{\frac{1}{\mu}} dt^2 + f(R,\mu)^{-\frac{1}{\mu}} dR^2 + (R + \tilde{r_0}(\mu+1))(R - \tilde{r_0}(\mu-1)) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2).$$

We will now focus only on the last term, corresponding to the angular part of the metric

$$\begin{aligned}
&(R + \tilde{r_0}(\mu+1))(R - \tilde{r_0}(\mu-1)) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2) \\
&(R + M + M\mu)(R - M - M\mu)) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2) \\
&(\tilde{R} + \tilde{M})(\tilde{R} - \tilde{M})) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2) \\
&(\tilde{R}^2 - \tilde{M}^2) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2),
\end{aligned}$$

where in the last two lines we have used a reparametrization $\tilde{R} = R + M$, $\tilde{M} = M\mu$ same as in [21], which we can do for large μ, since M is proportional to $1/\mu$ as $M = \sqrt{\frac{F}{\mu^2 - 1}} \propto \frac{1}{\mu}$, where F is a constant, so the term does not diverge on itself.
The simplified line element has been modified to

$$ds^2 = -f(R,\mu)^{\frac{1}{\mu}} dt^2 + f(R,\mu)^{-\frac{1}{\mu}} dR^2 + (\tilde{R}^2 - \tilde{M}^2) f(R,\mu)^{-\frac{1}{\mu}} (d\theta^2 + \sin^2\theta d\varphi^2).$$

Now, if we take the limit $\mu \to +\infty$, we have $f(R,\mu)^{\frac{1}{\mu}} \to 1$, so we simply get

$$ds^2 = -dt^2 + dR^2 + (\tilde{R}^2 - \tilde{M}^2)(d\theta^2 + \sin^2\theta d\varphi^2). \tag{2.34}$$

This is the ultrastatic limit of FJNW metric, since none of the spatial components of the metric depend explicitly on t. Note that there is no ambiguity now, the **limit is unique**.
Let us now focus on the original function $r(R)$ and show how the radial coordinates behave. We have

$$\begin{aligned}
r^2 &= \lim_{\mu \to +\infty} \frac{1}{4}[2R + 2M(\mu+1)]^{1+\frac{1}{\mu}} [2R - 2M(\mu-1)]^{1-\frac{1}{\mu}} \\
r^2 &= \lim_{\mu \to +\infty} \frac{1}{4}[2\tilde{R} + 2\tilde{M}]^{1+\frac{1}{\mu}} [2\tilde{R} - \tilde{M}]^{1-\frac{1}{\mu}} \\
r^2 &= \lim_{\mu \to +\infty} \frac{1}{4}[4\tilde{R}^2 - 4\tilde{M}^2] \\
r^2 &= \tilde{R}^2 - \tilde{M}^2.
\end{aligned}$$

We will rename $\tilde{R}$, $\tilde{M}$ to R and M respectively for simplicity. Thus

$$r^2 = R^2 - M^2. \tag{2.35}$$

So our metric becomes

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & (R^2 - M^2) & 0 \\ 0 & 0 & 0 & (R^2 - M^2)\sin^2(\theta) \end{pmatrix}. \tag{2.36}$$

And taking the inverse of this matrix, we have

$$g^{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{(R^2 - M^2)} & 0 \\ 0 & 0 & 0 & \frac{1}{(R^2 - M^2)\sin^2(\theta)} \end{pmatrix}. \tag{2.37}$$

Now we can solve the equation (1.32) using this metric. To do that we need to find out our expression (2.2) for this limit.

$$\phi_{\mu\to+\infty} = \lim_{\mu\to+\infty} \frac{A}{\mu} \ln\left|\frac{R-M}{R+M}\right| = \frac{1}{\sqrt{2}} \ln\left|\frac{R-M}{R+M}\right|, \tag{2.38}$$

since these constants can be related as[22] $\mu = \sqrt{1+2A^2}$. Computing the gradient of the potential, we have

$$\phi_{,R} = \frac{1}{\sqrt{2}} \frac{R+M}{R-M} \frac{2M}{(R+M)^2} = \frac{1}{\sqrt{2}} \frac{2M}{R^2 - M^2} = \frac{M\sqrt{2}}{R^2 - M^2}$$

$$\phi_{,\mu} = \left(0, \frac{M\sqrt{2}}{R^2 - M^2}, 0, 0\right), \tag{2.39}$$

so the Ricci tensor is easily

$$R_{\mu\nu} = \frac{4M^2}{(R^2 - M^2)^2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{2.40}$$

And the Ricci scalar

$$R = g^{\mu\nu} R_{\mu\nu} = g^{RR} R_{RR} = \frac{4M^2}{(R^2 - M^2)^2}, \tag{2.41}$$

So there exists a curvature singularity at $R = M$. From (2.35) we also know that this is still a single point in space. The singularity is time-like in this case.
Now, for Christoffel symbols, we can, for simplicity, do the computation in MAPLE, since we have illustrated the previous solution in detail. That leaves us with the following set of geodetic equations already simplified, the same way we previously

did without loss of generality, thus $\theta = \pi/2$ and $\dot{\theta} = 0$ at $t = 0$. The right hand side of the equation (1.32) can be written as

$$-\frac{q'}{1+q'\frac{1}{\sqrt{2}}\ln\left|\frac{R-M}{R+M}\right|}(u^R u^\nu + g^{R\nu})\frac{M\sqrt{2}}{R^2-M^2}$$
$$-\frac{q'}{\sqrt{2}+q'\ln\left|\frac{R-M}{R+M}\right|}(u^R u^\nu + g^{R\nu})\frac{2M}{R^2-M^2}.$$

We can introduce a term $C(R)$ analogical to the previous term B as

$$C(R) = \frac{q'}{\sqrt{2}+q'\ln\left|\frac{R-M}{R+M}\right|}\frac{2M}{R^2-M^2}, \tag{2.42}$$

so the right hand side becomes

$$-C(R)(u^R u^\nu + g^{R\nu}).$$

For the general set of equations, we have

$$\ddot{t} = -C(R)\dot{R}\dot{t} \tag{2.43}$$
$$\ddot{R} - R\dot{\varphi}^2 = -C(R)(1+\dot{R}^2) \tag{2.44}$$
$$\ddot{\theta} = 0 \tag{2.45}$$
$$\ddot{\varphi} - \frac{2R\dot{R}\dot{\varphi}}{(M^2-R^2)} = -C(R)\dot{\varphi}\dot{R}. \tag{2.46}$$

If we were to again apply the condition that $\dot{\varphi} = 0$ as an initial condition, then we would end up with

$$\ddot{t} = -C(R)\dot{R}\dot{t}$$
$$\ddot{R} = -C(R)(1+\dot{R}^2)$$
$$\ddot{\theta} = 0$$
$$\ddot{\varphi} = 0.$$

Now, for the charge horizon, the following condition $1 + q'\phi = 0$ must hold. For that to be true we get the following expression

$$R_C = \frac{M}{\tanh\left(\frac{\sqrt{2}}{2q'}\right)}. \tag{2.47}$$

Now we can again plot it as a function of q', so we get the following figure.

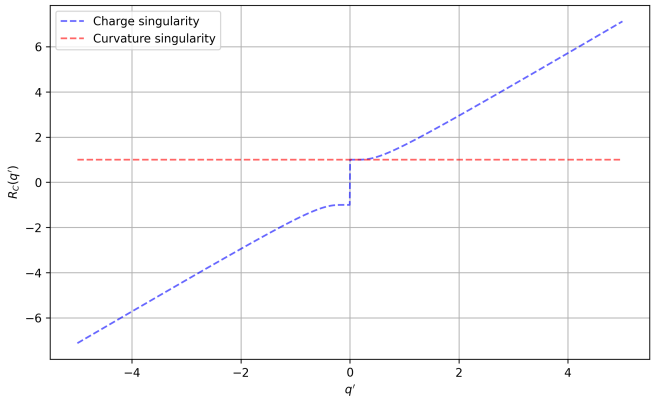


Figure 2.14 Critical radius R_C as a function of charge q'. We have set $M = 1$.

We thus have the same behaviour - the charge horizon is enclosing the curvature singularity only if the charge is positive.
Now we can show for which values the second derivative becomes positive. Taking the more general solution, we have

$$\ddot{R} = -\frac{q'}{\sqrt{2} + q' \ln\left|\frac{R-M}{R+M}\right|}(1 + \dot{R}^2)\frac{2M}{R^2 - M^2} + R\dot{\varphi}^2. \tag{2.48}$$

The only term that can change signs is the first one in the denominator. We will define a new function as the denominator, i.e. $D(R)$.
So for the denominator, we get the following plot for $D(R)$ for negative q'.

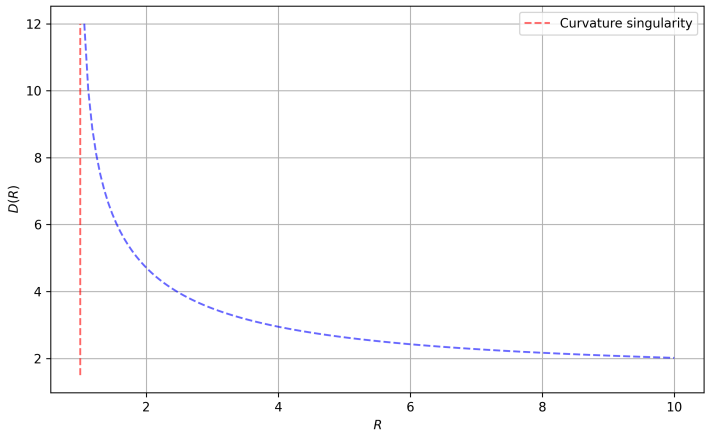


Figure 2.15 Denominator term $D(R)$ for negative q'. We have set $M = 1$ and $q' = -3$.

So for this charge, the denominator doesn't change signs, thus the whole first term is positive throughout the whole evolution of the system. The negative charge is therefore always repelled from the centre.
And for positive q' of the same magnitude, we get the following figure.

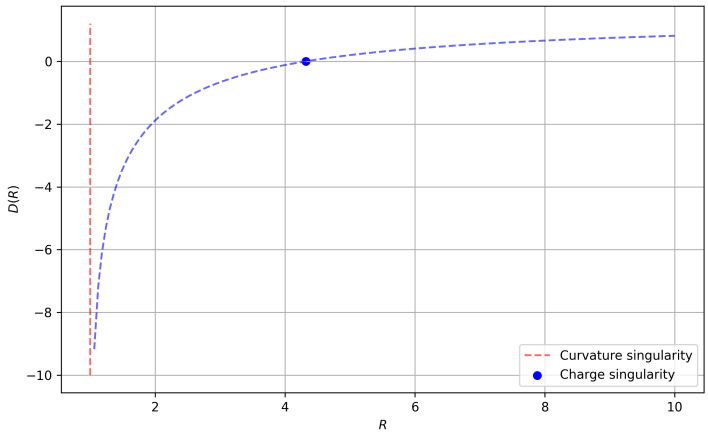


Figure 2.16 Denominator term $D(R)$ for positive q'. We have set $M = 1$ and $q' = 3$.

So it changes signs when the particle 'crosses' the charge singularity. That means if we are inside this sphere, the second derivative is positive. If we are outside, the sign of the first term changes and we again have a competition between the two terms which decides the sign of the second derivative.
It can be immediately seen from (2.48), the second derivative is always non-negative if the particle possesses no charge.
Now, again, plotting in the usual Cartesian coordinates, we have the following figure after also employing the repulsive radius sketch.

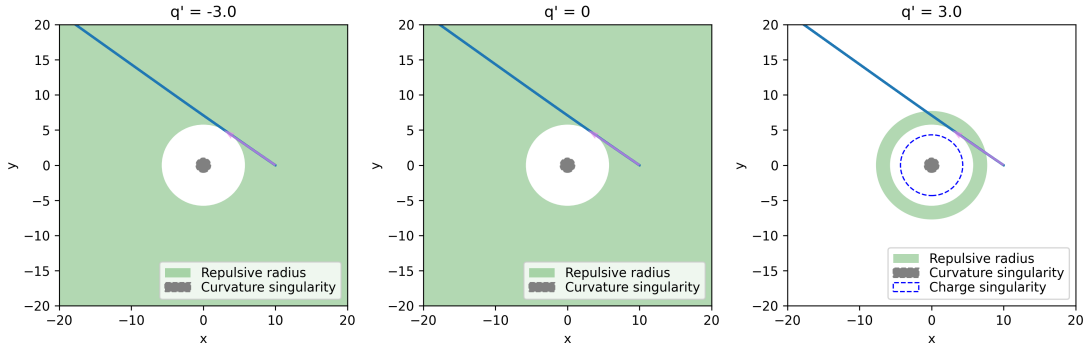


Figure 2.17 Geodesics in the ultrastatic limit of FJNW spacetime outside the naked singularity with $M = 1$.

For this geodesic, the particle started its motion outside the charge singularity, so for the negative charge it is repelled throughout the whole evolution of the system. For the particle with positive charge, we see that it is affected by the resulting repulsive force when near the centre. We see that the repulsive radius is consistent with the discussion of term $D(R)$.
We can also try to start inside the charge singularity, but since the term $1/D(R)$ diverges near this sphere, it is impossible to control the normalization of four-velocity, so we cannot be truly sure about the statement that the particle would actually reach it or be repelled near this sphere. Starting inside the sphere of the charge singularity, we have obtained the following figure.

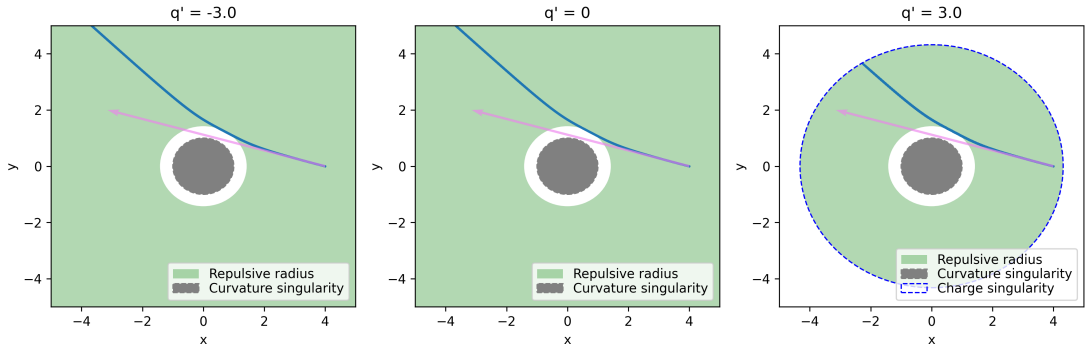


Figure 2.18 Geodesics in the ultrastatic limit of FJNW spacetime inside the charge horizon with $M = 1$.

This shows that, once more, in the uncharged case, the particle is always repelled. In the negatively charged case, the particle is repelled in the entirety of its motion. And finally for the positively charged case, the particle is repelled, until it reaches the charge singularity.

2.3.1 Circular orbits in the ultrastatic limit

If we once again impose the condition that $\dot{R} = 0$, we get the following set of equations

$$\begin{aligned} \ddot{t} &= 0 \\ R\dot{\varphi}^2 &= C(R) \\ \ddot{\theta} &= 0 \\ \ddot{\varphi} &= 0. \end{aligned}$$

Since R is constant in this case, we have the explicit equation for $\dot{\varphi}$

$$\dot{\varphi} = \sqrt{\frac{C(R)}{R}}. \tag{2.49}$$

To visualize this, we need to set the initial value for either $\dot{\varphi}$ or $\dot{t}$, and calculate the other through normalization of four-velocity, since we consider both $\dot{\theta}$ and $\dot{R}$ to be zero. However, we see immediately that the only non-trivial solution is for $q' > 0$. For $q' < 0$, the term $C(R) < 0$ always, so the equation has no real solutions. And for the uncharged particle, the solution is trivially constant. Plot in Cartesian coordinates is presented below for the positive charge.

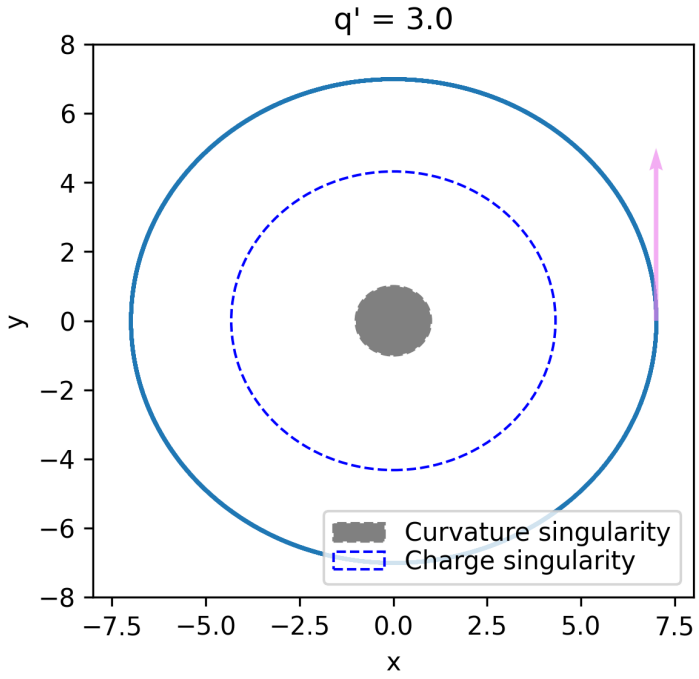


Figure 2.19 Circular orbits in the ultrastatic limit of FJNW spacetime with $M = 1$.

In the ultrastatic limit circular orbits exist only for a positively charged particle.

2.4 Effective potential

Using our derived expression for the action, we have

$$S = -m_0 \int (1 + q'\phi) d\tau. \tag{2.50}$$

And since

$$S = \int \mathcal{L} d\tau,$$

we can treat the factor of -1 as the normalization of four-velocity, which leaves us with

$$\mathcal{L} = \frac{1}{2} m_0 (1 + q'\phi) g_{\mu\nu} u^\mu u^\nu, \tag{2.51}$$

where the factor $1/2$ is just a rescaling by convention and doesn't change the equations of motion. We can also see this choice for the Lagrangian from the one of a free particle with no scalar field $\mathcal{L} = -m_0$, which (again not considering the factor $1/2$) corresponds to the form $\mathcal{L} = \frac{1}{2} m_0 g_{\mu\nu} u^\mu u^\nu$.
If we take the FJNW metric, then two coordinates are cyclic, namely t and φ. This corresponds to two integrals of motion, the conservation of energy E and the conservation of angular momentum L.

So, imposing a diagonal metric, we have

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial u^t} &= \frac{1}{2} m_0 (1 + q'\phi) g_{tt} u^t \\
\frac{\partial \mathcal{L}}{\partial u^r} &= \frac{1}{2} m_0 (1 + q'\phi) g_{rr} u^r \\
\frac{\partial \mathcal{L}}{\partial u^\theta} &= \frac{1}{2} m_0 (1 + q'\phi) g_{\theta\theta} u^\theta \\
\frac{\partial \mathcal{L}}{\partial u^\varphi} &= \frac{1}{2} m_0 (1 + q'\phi) g_{\varphi\varphi} u^\varphi.
\end{aligned}$$

However we need to check if these still correspond to the values we know, namely energy E and angular momentum L. These should correspond to the t and φ indices of the four-velocities derivatives respectively.
To explore this, we begin by examining the mass correspondence using the definition of angular momentum.

Mass correspondence in FJNW

We have the relation for the φ-component of the angular momentum[12]

$$p^\varphi = \frac{p_\varphi}{g_{\varphi\varphi}} = \frac{L}{g_{\varphi\varphi}}, \tag{2.52}$$

where p^φ is the contravariant component of the angular part of the four-momentum and L is the angular momentum. The contravariant component of the four-momentum is defined as

$$p^\varphi = m'\dot{\varphi}, \tag{2.53}$$

where m' has a meaning of mass, whose explicit form we will see further on.
And from here we can set again $\theta = \pi/2$ to get

$$\begin{aligned}
&p^\varphi = \frac{L}{r^2 f^{1-n}} \\
&p^\varphi r^2 f^{1-n} = L \\
&m'\dot{\varphi} r^2 f^{1-n} = L \\
&\frac{\mathrm{d}}{\mathrm{d}\tau}(m'\dot{\varphi} r^2 f^{1-n}) = \frac{\mathrm{d}L}{\mathrm{d}\tau} \\
&\dot{m'}\dot{\varphi} r^2 f^{1-n} + m'\ddot{\varphi} r^2 f^{1-n} + m'\dot{\varphi} 2r\dot{r} f^{1-n} + m'\dot{\varphi}\dot{r} r^2 (1-n) f^{-n} \frac{2M}{nr^2} = 0 \\
&\ddot{\varphi} + 2\frac{1}{r}\dot{r}\dot{\varphi} + \dot{\varphi}\dot{r}(1-n) f^{-1} \frac{2M}{nr^2} + \frac{1}{m'}\dot{m'}\dot{\varphi} = 0 \\
&\ddot{\varphi} + \frac{2M\dot{r}\dot{\varphi}}{fr^2}\left(-1 + \frac{r}{M} - \frac{1}{n}\right) + \frac{1}{m'}\dot{m'}\dot{\varphi} = 0.
\end{aligned}$$

Now this equation has to be the same equation as we had for the φ-component in our equations of motion (2.26). For that to be true, the following relation must hold

$$\begin{aligned}
\frac{1}{m'}\dot{m'}\dot{\varphi} &= \frac{M\dot{r}\dot{\varphi}B}{fr^2} \\
\frac{1}{m'}\dot{m'} &= \frac{M\dot{r}}{fr^2}\frac{q'\sqrt{1-n^2}}{n(q'\phi+1)}.
\end{aligned}$$

We see that m' must depend on ϕ. Therefore a reasonable assumption is to use the dynamic mass $m' = m_0(1 + q'\phi)$. After checking the solution, we see that this does in fact hold, since

$$\dot{\phi} = \frac{q'\sqrt{1-n^2}}{fr^2n} M\dot{r}.$$

The rest masses m_0 cancel out, since they are invariant. The previous relation can be fulfilled for any constant $m' = c(1 + q'\phi)$, however we can take it to be the rest mass.
Another way to show the form of m' is to start from the equation of motion (1.38) that we got from using the dynamical mass in our Lagrangian. For the equation of motion using the four-momentum, we have

$$\frac{Dp_\mu}{d\tau} = -m'_{,\mu}. \tag{2.54}$$

And now, noting that all the right hand sides must be zero, with an exception $\mu = r$, since we have two integrals of motion t, φ and are considering only planar motion $\dot{\theta} = 0$, we have

$$\frac{Dp_r}{d\tau} = -m'_{,r},$$

so $m' = m'(r)$. Using the chain rule and plugging in the expression for p_r, we have

$$\begin{aligned}
\frac{D(m'u_r)}{d\tau} &= -m'_{,r} \\
\frac{D(m')}{d\tau}u_r + m'\frac{D(u_r)}{d\tau} &= -m'_{,r} \\
\frac{D(u_r)}{d\tau} &= \frac{1}{m'}(-m'_{,r} - \frac{D(m')}{d\tau}u_r) \\
\frac{D(u_r)}{d\tau} &= \frac{1}{m'}(-m'_{,r} - m'_{,r}\dot{r}u_r) \\
\frac{D(u_r)}{d\tau} &= -\frac{m'_{,r}}{m'}(1 + g_{rr}\dot{r}^2) \\
\frac{D(u_r)}{d\tau}g^{rr} &= -\frac{m'_{,r}}{m'}(g^{rr} + \dot{r}^2).
\end{aligned}$$

But this is a very similar expression to the one we derived earlier in (1.32). It reduces to that solution only if

$$m' = c(1 + q'\phi), \tag{2.55}$$

where c is a constant. That means if we take $c = m_0$, we would have $m' = \mathcal{M}$. For clarification this was just a demonstration, since as we have set the mass m' in (2.54), the identity $m' = \mathcal{M}$ follows immediately up to a constant factor. However it does justify our initial form of the four-momentum.

Mass correspondence in the ultrastatic limit

Employing the same approach as above, we have (setting $\theta = \pi/2$)

$$\begin{aligned}
&p^{\varphi} = \frac{L}{(R^2 - M^2)} \\
&p^{\varphi}(R^2 - M^2) = L \\
&\frac{\mathrm{d}}{\mathrm{d}\tau}(m'\dot{\varphi}(R^2 - M^2)) = \frac{\mathrm{d}L}{\mathrm{d}\tau} \\
&\dot{m'}\dot{\varphi}(R^2 - M^2) + m'\ddot{\varphi}(R^2 - M^2) + m'\dot{\varphi}2R\dot{R} = 0 \\
&\ddot{\varphi}(R^2 - M^2) + \dot{\varphi}2R\dot{R} + \frac{1}{m'}\dot{m'}\dot{\varphi}(R^2 - M^2) = 0 \\
&\ddot{\varphi} + \frac{\dot{\varphi}2R\dot{R}}{(R^2 - M^2)} + \frac{1}{m'}\dot{m'}\dot{\varphi} = 0,
\end{aligned}$$

which has to be the same equation as (2.46), thus

$$\begin{aligned}
\frac{1}{m'}\dot{m'}\dot{\varphi} &= -C(R)\dot{\varphi}\dot{R} \\
\frac{1}{m'}\dot{m'} &= \frac{q'M\sqrt{2}}{(1 + q'\phi)(R^2 - M^2)}\dot{R}.
\end{aligned}$$

Again, assuming $m' = m_0(1 + q'\phi)$ fulfils this relation, since

$$\dot{\phi} = \frac{M\sqrt{2}}{R^2 - M^2}\dot{R}.$$

2.4.1 Conserved quantities

For the effective potential, we will need to have the explicit equation for our integrals of motion. To do that, we will follow a similar approach as in[12] and check the limit $r \to +\infty$, respectively $R \to +\infty$ for FJNW, respectively its ultrastatic limit. In both these limits, we know that $\phi \to 0$.

Integrals of motion in FJNW

Starting with FJNW, we have for $\dot{t}$

$$\dot{t} = \frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{dt}{\sqrt{-g_{\mu\nu}dx^{\mu}dx^{\nu}}} = \frac{1}{\sqrt{-g_{\mu\nu}\dfrac{\mathrm{d}x^{\mu}}{\mathrm{d}t}\dfrac{\mathrm{d}x^{\nu}}{\mathrm{d}t}}} = \frac{1}{\sqrt{f^n - g_{ij}\dfrac{\mathrm{d}x^{i}}{\mathrm{d}t}\dfrac{\mathrm{d}x^{j}}{\mathrm{d}t}}} = \frac{1}{\sqrt{f^n - v^2}},$$

where v is the three-velocity. Now we can put this into our expression for E the Lagrangian derivative

$$\frac{1}{2}m_0(1 + q'\phi)(-f^n)\frac{1}{\sqrt{f^n - v^2}},$$

and taking the limit of $r \to +\infty$, we immediately have

$$\frac{1}{2}m_0(-1)\frac{1}{\sqrt{1 - v^2}} = -\frac{1}{2}m_0\gamma = -\frac{1}{2}E.$$

So this is just the rescaled particle's energy at infinity.
And for $\dot{\varphi}$, we use the chain rule to get

$$\dot{\varphi} = \frac{\mathrm{d}\varphi}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\tau} = v^{\varphi}\dot{t},$$

where v^{φ} represents the angular part of the three-velocity.
Now for the φ derivative of the Lagrangian, we have

$$\frac{1}{2}m_0(1+q'\phi)(r^2\sin^2(\theta)f^{1-n})\dot{\varphi} = \frac{1}{2}m_0(1+q'\phi)(r^2\sin^2(\theta)f^{1-n})v^{\varphi}\frac{1}{\sqrt{f^n - v^2}},$$

which, again, after taking the same limit, reduces to

$$\frac{1}{2}m_0 r^2\sin^2(\theta)v^{\varphi}\gamma = \frac{1}{2}E\lim_{r\to+\infty}(r^2\sin^2(\theta)v^{\varphi}) = \frac{1}{2}L.$$

Integrals of motion in the ultrastatic limit

Continuing with the ultrastatic limit, we have for $\dot{t}$

$$\dot{t} = \frac{1}{\sqrt{1-v^2}}.$$

Now again in the Lagrangian derivative, this is simply

$$\frac{1}{2}m_0(1+q'\phi)(-1)\frac{1}{\sqrt{1-v^2}},$$

which in the limit $R\to+\infty$ gives

$$-\frac{1}{2}m_0\frac{1}{\sqrt{1-v^2}} = -\frac{1}{2}m_0\gamma = -\frac{1}{2}E.$$

And for the φ part of the derivative of the Lagrangian, we get

$$\frac{1}{2}m_0(1+q'\phi)(R^2-M^2)\sin^2(\theta)\dot{\varphi} = \frac{1}{2}m_0(1+q'\phi)(R^2-M^2)\sin^2(\theta)v^{\varphi}\frac{1}{\sqrt{1-v^2}},$$

which in the same limit gives

$$\frac{1}{2}m_0R^2\left(1-\frac{M^2}{R^2}\right)\sin^2(\theta)v^{\varphi}\gamma = \frac{1}{2}E\lim_{R\to+\infty}R^2v^{\varphi}\sin^2(\theta) = \frac{1}{2}L.$$

Employing normalization of four-velocity

So these are the correct corresponding integrals of motion. Explicitly for them we have

$$E = -m_0(1+q'\phi)g_{tt}u^t \tag{2.56}$$
$$L = -m_0(1+q'\phi)g_{\varphi\varphi}u^{\varphi} \tag{2.57}$$

Now, focusing on effective potential, we have

$$\frac{E^2}{g_{tt}} = (m_0(1+q'\phi))^2 g_{tt}u^t u^t$$
$$\frac{L^2}{g_{\varphi\varphi}} = (m_0(1+q'\phi))^2 g_{\varphi\varphi}u^{\varphi}u^{\varphi}.$$

Next, we will use the normalization of four-velocity and use the form for energy E to get the dependency of the motion in the radial coordinate.

$$\begin{aligned}
&g_{\mu\nu}u^\mu u^\nu = -1 \\
&g_{tt}\dot{t}^2 + g_{rr}\dot{r}^2 + g_{\theta\theta}\dot{\theta}^2 + g_{\varphi\varphi}\dot{\varphi}^2 = -1 \\
&\frac{1}{(m_0(1+q'\phi))^2}\left(\frac{E^2}{g_{tt}} + \frac{L^2}{g_{\varphi\varphi}}\right) + g_{rr}\dot{r}^2 + g_{\theta\theta}\dot{\theta}^2 = -1
\end{aligned}$$

We can now set $\theta = \pi/2$ and $\dot{\theta} = 0$ (which we will be setting implicitly further on) and get the expression for $\dot{r}$

$$\dot{r}^2 = \frac{1}{g_{rr}}\left(-1 - \frac{1}{(m_0(1+q'\phi))^2}\left(\frac{E^2}{g_{tt}} + \frac{L^2}{g_{\varphi\varphi}}\right)\right).$$

2.4.2 Effective potential in FJNW

Using the metric for the FJNW spacetime, we have

$$\begin{aligned}
E &= \frac{\partial \mathcal{L}}{\partial \dot{t}} = m_0(1+q'\phi)f^n\dot{t} \\
L &= \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = m_0(1+q'\phi)r^2 f^{1-n}\dot{\varphi}.
\end{aligned}$$

We can use these in the normalization of four-velocity

$$\begin{aligned}
\dot{r}^2 &= f^n\left(-1 - \frac{1}{(m_0(1+q'\phi))^2}\left(-\frac{E^2}{f^n} + \frac{L^2}{r^2 f^{1-n}}\right)\right) \\
\dot{r}^2 &= -f^n + \frac{1}{(m_0(1+q'\phi))^2}\left(E^2 - \frac{L^2 f^{2n-1}}{r^2}\right) \\
\dot{r}^2 &= \frac{1}{(m_0(1+q'\phi))^2}E^2 - \left(\frac{1}{(m_0(1+q'\phi))^2}\frac{L^2 f^{2n-1}}{r^2} + f^n\right) \\
\dot{r}^2 &= \frac{1}{(m_0(1+q'\phi))^2}E^2 - V_{eff},
\end{aligned}$$

where we have defined the effective potential V_{eff} as

$$V_{eff} = \frac{1}{(m_0(1+q'\phi))^2}\frac{L^2 f^{2n-1}}{r^2} + f^n. \tag{2.58}$$

But now, instead of the usual constraint $V_{eff} \leq E^2$, we need to fulfil the condition

$$V_{eff} \leq \frac{1}{(m_0(1+q'\phi))^2}E^2 =: E_{mod}, \tag{2.59}$$

for motion to be possible. we have denoted the right hand side as E_{mod} in the following figures for simplicity. This reduces to the usual constraint mentioned

above, if we set $q' = 0$ and multiply the inequality by m_0^2, thus rescaling both the effective potential and the radial part of four-velocity by this value.
If we multiply both these factors by the term $(m_0(1 + q'\phi))^2$, we get the constant energy E and the effective potential multiplied with this term does not diverge at the charge horizon. If then these two factors are equal, we immediately have $\dot{r} = 0$, signifying a turning point. If that point is the minimum, respectively maximum of the effective potential multiplied by the factor, then these points represent stable, respectively unstable circular orbits.
From (2.58), we can now see how the limit for a particle reaching singularity using the effective potential behaves. Taking $n > 0.5$ and a charged particle, we have

$$\lim_{r \to \frac{2M}{n}} V_{eff} = 0,$$

since $f \to 0$. For an uncharged particle, the limit does not change.
However if we take $n = 0.5$, then the term f^{2n-1} is undefined, where we have the corresponding limit as from the original formulation of the FJNW spacetime. Nevertheless, considering this value to be finite, we can obtain for the same limit in the case of a charged particle the following value of the effective potential

$$\lim_{r \to \frac{2M}{n}^+} V_{eff} = 0,$$

the same as above. However for an uncharged particle, we would have, taking the previously discussed ambiguous limit to be 1,

$$\lim_{r \to \frac{2M}{n}^+} V_{eff} = \frac{L^2 n^2}{4 m_0^2 M^2}. \tag{2.60}$$

And from (2.58), we now see how important was the fact that we are considering $n \geq 0.5$. For $n < 0.5$ we would have for a charged particle

$$\lim_{r \to \frac{2M}{n}^+} V_{eff} = +\infty,$$

while the energy term E_{mod} is still finite, so it could not be reached by the particle. The same would hold also for the uncharged particle.
We can still recover the effective potential for Schwarzschild if we set $n = 1$ and $q' = 0$, thus

$$V_{eff,S} = \left(1 - \frac{2M}{r}\right)\left(\frac{L^2}{m_0 r^2} + 1\right).$$

Below, we show the plot for both sides of (2.59) as a function of r.

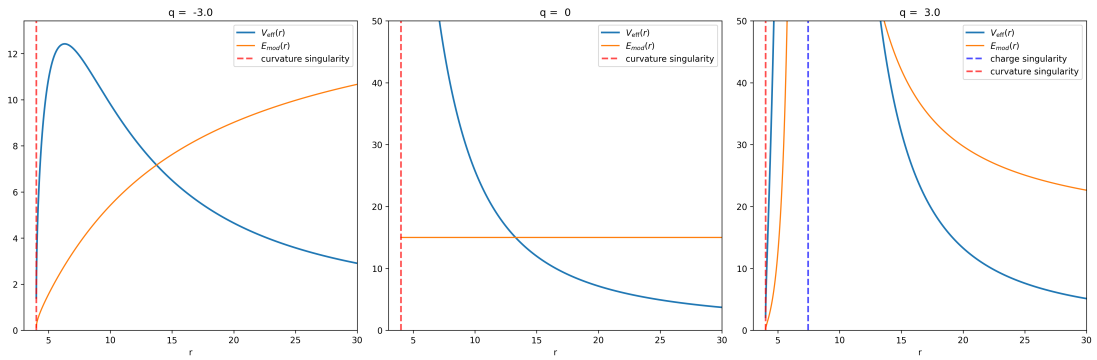


Figure 2.20 Effective potential V_{eff} and modified energy factor E_{mod} in FJNW spacetime with $m_0 = 0.1$, $M = 1$, $n = 0.5$, $E = 15$ and $L = 5M$. Energy E and the angular momentum L are set to the same value in all cases.

We can see that the effective potential spikes for $r = r_C$, but so does the E_{mod}. We can see that there is a peak in the effective potential near the curvature singularity for the negatively charged particle. This can be linked to the repulsive radius illustrated in Fig.2.3, since near the curvature singularity this particle is repelled. It can only reach the curvature singularity if it has sufficient energy when inside this radius.
We know that for the uncharged particle the charge singularity coincides with the curvature singularity from Fig.2.1.
Introducing a smaller value of m_0, we have to plot the effective potential and the modified energy factor in logarithmic scales.

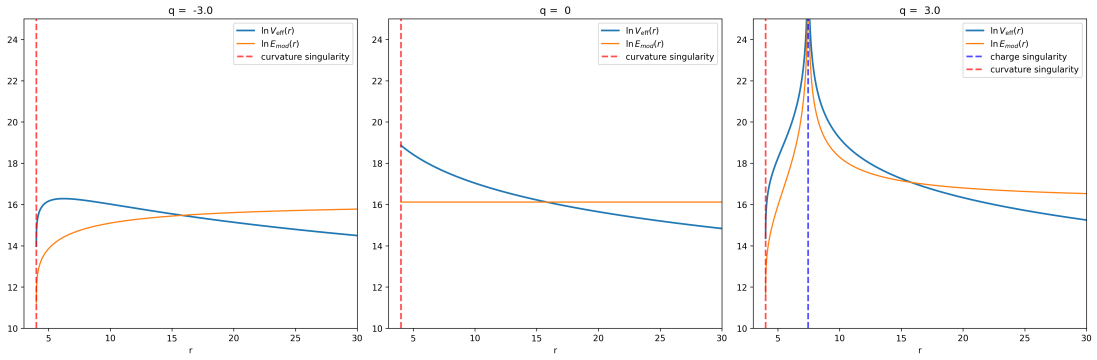


Figure 2.21 Logarithm of the effective potential V_{eff} and modified energy factor E_{mod} in FJNW spacetime with $m_0 = 10^{-4}$, $M = 1$, $n = 0.5$, $E = 15$ and $L = 5M$. Energy E and the angular momentum L are set to the same value in all cases.

And for a different parameter $n = 0.8$, we obtain the following figure.

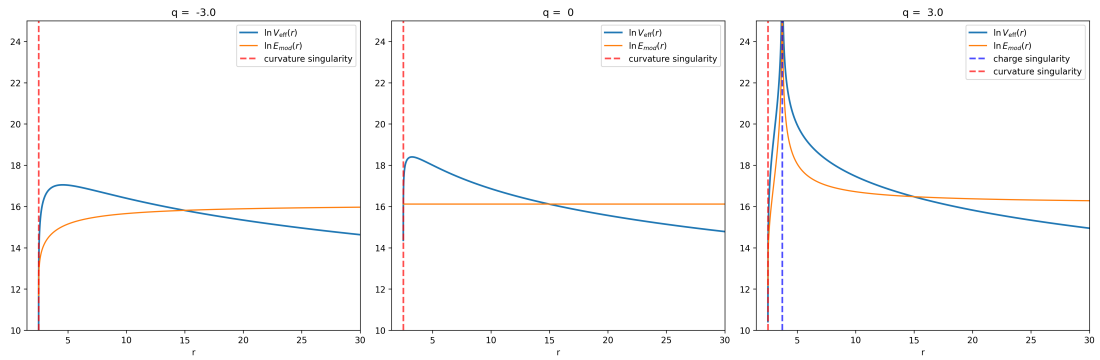


Figure 2.22 Logarithm of the effective potential V_{eff} and modified energy factor E_{mod} in FJNW spacetime with $m_0 = 10^{-4}$, $M = 1$, $n = 0.8$ and $L = 5M$. Energy E is set to the same value in all cases.

We see that the motion looks very similar with the exception of $q' = 0$, where near the singularity now the effective potential is approaching 0 in the limit. In the case for $n = 0.5$, this was not the case as we had proven previously in 2.60. If we multiply both V_{eff} and E_{mod} by factor $(m_0(1 + q'\phi))^2$ we get a constant energy factor, and the following figure for larger charges.

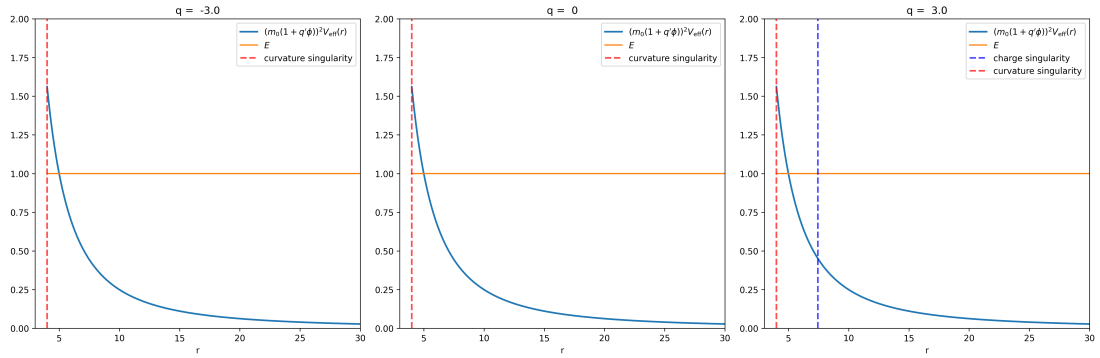


Figure 2.23 Effective potential V_{eff} multiplied by $(m_0(1+q'\phi))^2$ and constant energy E in FJNW. Constants are set to $m_0 = 10^{-4}, M = 1$, $n = 0.5$, $E = 1$ and $L = 5M$.

We see that the possible motion is very similar for all of these charge values. And near the charge singularity, there is no defect for the positive charge. However, that is due to the fact that near this radius we are effectively multiplying the whole equation by zero.
The graphs look very similar, since in(2.58) the only term dependent on charge is $f^n(m_0(1 + q'\phi))^2$, which is quadratic in m_0.
To visualize the difference, we need to consider larger masses of the particle. We obtain the following possible areas of motion.

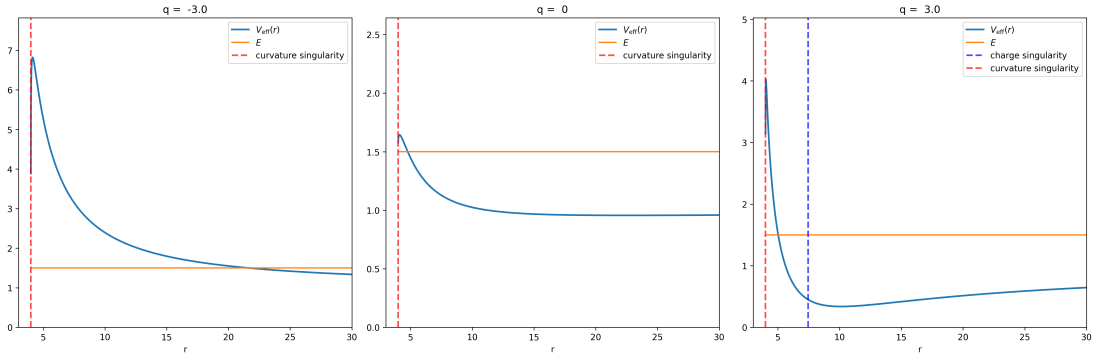


Figure 2.24 Effective potential V_{eff} and a constant energy E in FJNW both multiplied by $(m_0(1+q'\phi))^2$. Constants are set to $m_0 = 1, M = 1$, $n = 0.5$, $E = 1.5$ and $L = 5M$. Energy E is set to the same value in all cases.

For large particle masses and large positive charges, there exist stable circular orbits outside the charge horizon. There are also unstable circular orbits near the curvature singularity for all possible charges.

2.4.3 Effective potential in the ultrastatic limit

And for the ultrastatic limit, we have

$$E = \frac{\partial \mathcal{L}}{\partial \dot{t}} = m_0(1+q'\phi)\dot{t}$$
$$L = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = m_0(1+q'\phi)(R^2 - M^2)\dot{\varphi}.$$

And, again, using the normalization of four-velocity, we have

$$\dot{r}^2 = \left(-1 - \frac{1}{(m_0(1+q'\phi))^2}\left(-E^2 + \frac{L^2}{R^2 - M^2}\right)\right)$$
$$\dot{r}^2 = \frac{1}{(m_0(1+q'\phi))^2}E^2 - \left(\frac{1}{(m_0(1+q'\phi))^2}\frac{L^2}{R^2 - M^2} + 1\right)$$
$$\dot{r}^2 = \frac{1}{(m_0(1+q'\phi))^2}E^2 - V_{eff},$$

where we have now defined the effective potential to be

$$V_{eff} = \frac{1}{(m_0(1+q'\phi))^2}\frac{L^2}{R^2 - M^2} + 1. \tag{2.61}$$

Once more, taking the limit of $R \to M^+$, we have the following for both the charged and uncharged particles

$$\lim_{r \to M^+} V_{eff} \cdot (m_0(1+q'\phi))^2 = +\infty.$$

This simply means that the **particle cannot reach the curvature singularity** in this spacetime, because it would need an infinite amount of energy. We had the same case previously, when we considered $n < 0.5$.
And we have the same constraint for the motion as we had in FJNW. Now, again, we can visualize these constraints in the same way as before.

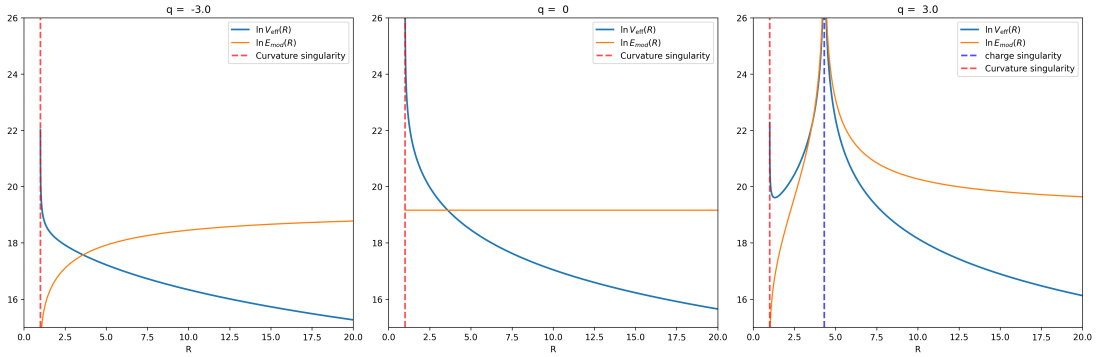


Figure 2.25 Logarithm of the effective potential V_{eff} and logarithm of the modified energy factor E_{mod} in the ultrastatic limit of FJNW spacetime with $m_0 = 10^{-4}$, $M = 1$, $E = 1.2$ and $L = 5M$.

We see again that for the charge singularity both the effective potential and the modified energy factor diverge. However that does not necessarily imply any abnormal behaviour near this radius.
We know that both the negatively charged particle and the uncharged particle are repelled from the centre during the entire motion. From Fig.2.25 we can see that the resulting repulsive force becomes stronger with decreasing distance.
However for the positively charged particle, the curvature singularity also cannot be reached, even when inside the charge horizon. Nevertheless, there exist stable circular orbits for the positively charged particle, when inside the charge horizon. Once more, we consider a particle with the same charge, but a higher mass.

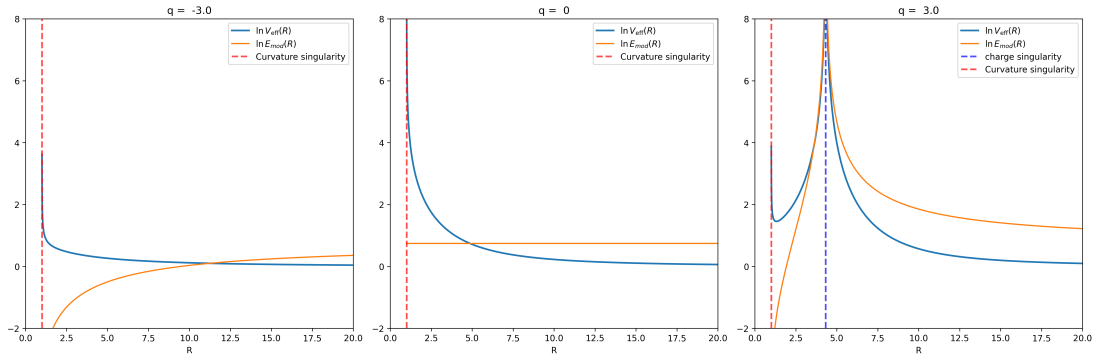


Figure 2.26 Logarithm of the effective potential V_{eff} and logarithm of the modified energy factor E_{mod} in the ultrastatic limit of FJNW spacetime with $m_0 = 1$, $M = 1$, $E = 1.2$ $L = 5M$.

We see that the effective potential looks almost the same as in Fig.2.25, only it has a gentler slope as we would expect, since the particle's acceleration is now less affected by external forces.
And, multiplying by the same factor as before $(m_0(1 + q'\phi))^2$, we get also the following figures for $m_0 = 10^{-4}$ and $m_0 = 1$ respectively.

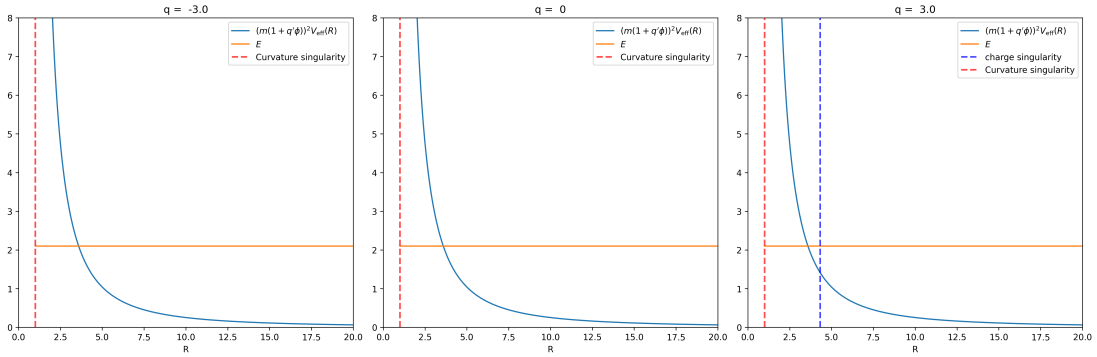


Figure 2.27 Effective potential multiplied by a factor $(m_0(1+q'\phi))^2 V_{eff}$ and constant energy E in the ultrastatic limit of FJNW spacetime with $m_0 = 10^{-4}$ $M = 1$, $E = 2.1$ $L = 5M$.

Once more, the plots look very similar for every value of the charge. Nevertheless we are still multiplying by zero near the charge singularity, so we cannot be sure of its appearance near this radius.
Again, plotting for the particle with a higher mass, we have the following behaviour.

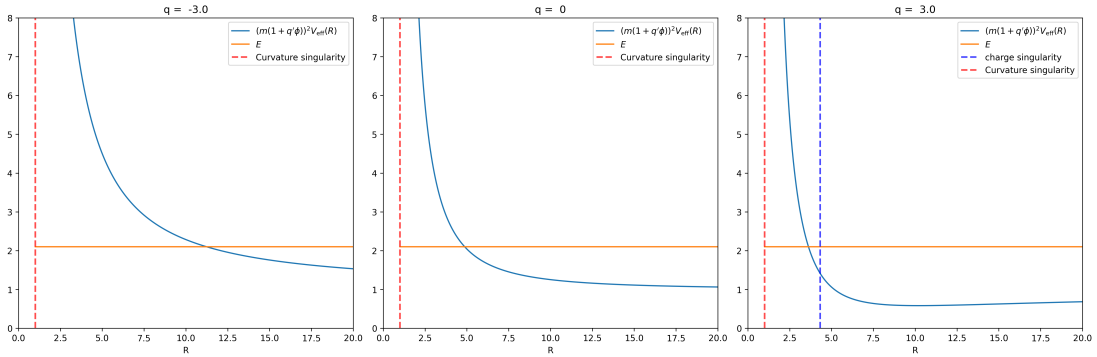


Figure 2.28 Effective potential multiplied by a factor $(m_0(1+q'\phi))^2 V_{eff}$ and constant energy E in the ultrastatic limit of FJNW spacetime with $m_0 = 1$ $M = 1$, $E = 2.1$ and $L = 5M$.

Here we can see that the effective potential multiplied by the factor does have (for an extremely large particle mass carrying positive charge) a global minimum, thus signifying stable circular orbits. This complies with our analysis of the equation(2.49) resulting in the motion illustrated in Fig.2.19.

Conclusion

In this work, we have explored the motion of a scalar massive particle in a gravitating scalar massless spherically symmetric field, namely the FJNW spacetime and its ultrastatic limit. We have derived the Lagrangian for a massive particle in a massless scalar field using Hamilton's principle and d'Alembert's principle. After deriving the Lagrangian, we compared and verified our form with other published papers. Next, we have derived the equations of motion from this Lagrangian by applying the principle of least action.
Furthermore we examined the key features of the FJNW spacetime, both in its original formulation and in its reparametrized form. Using the equations of motion we employed the reparametrized solution to derive an explicit form for further study. We have compared our results with those presented in a related study[18], whose main focus was on a different topic than ours. To verify these results, we also derived the same form by using `MAPLE` for computing the geodesic equations. In these equations, we discovered a sphere of a specific radius for which the right hand side of the equations of motion diverges in the radial component. For simplicity, we named this sphere as the charge horizon. This is also linked to the concept of dynamical mass, meaning that on this sphere, the dynamical mass is zero.
We then performed a numerical calculation of these equations in Python, using the function `solve_ivp` from the Scipy library. We have also found that there is a spherical shell with an inner radius, inside which the direction of the force points radially outward from the centre. We compared the whole system evolution to exterior Schwarzschild motion, which we obtained simply by setting the charge to zero and adjusting a parameter in the reparametrized form $n = 1$. Thus we observed that the charge of the particle truly corresponds to a modification of this spherical sector.
We then explored the circular orbits in FJNW, which differ considerably for varying particle charge. In the specific choice of initial values, we discovered that the positive charge corresponds to an attractive force towards the centre.
In addition to the FJNW metric, we computed the ultrastatic limit of the original formulation. We again explored the main aspects and found out that there is still the charge horizon, however with a different form. We have solved the equations of motion, illustrated the final solution, using `MAPLE`, and studied the behaviour of the spherical repulsive sector. The behaviour of this sector changed for the particle possessing positive charge when it crossed the charge horizon. Noting that the repulsive sector is also in Schwarzschild, there it corresponds to the centrifugal barrier term.
Analysing the ultrastatic limit, we found out that there are stable circular orbits and derived equations of motion for them. We found that these can exist only for a particle carrying positive charge.
Finally, we have calculated the effective potential for both spacetimes. This was important especially in determining the behaviour of the charge singularity. Noting that there are two conserved quantities in both these spacetimes, namely energy and angular momentum, we found that the charge singularity does not represent an unreachable sphere, but it can be crossed with sufficient energy. That

means that the curvature singularity is truly naked, because an observer may escape to infinity.
In the effective potential analysis, we have in some cases used extremely large particle masses to illustrate the motion. While we still treated these objects as test particles, this is a poor approximation for masses comparable to those of the singularity/black hole. Similarly, introducing a scalar charge of similar magnitude as we have in some cases for a test particle is not valid, as such a particle necessarily alters the scalar field through its own charge, violating the assumption of a fixed background.
However there remains another question with a more underlying theoretical aspect, being, what does the charge horizon in fact represent, since the geodesic equation at that radius breaks down, nevertheless as we have seen from the effective potential, that point in space can be reached, and crossed by the particle, implying that it should not be a pathology in this spacetime.
Another issue that is hinted at throughout our whole work is the physical meaning of the charge. We have seen that in some special cases its physical interpretation depends on the distance from the centre. Namely, for a particle with negative charge, the force is always repulsive in the ultrastatic limit. However that is not the case in FJNW. There, even a positively charged particle can experience a resulting repulsive force when near the curvature singularity.

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A Attachments

A.1 Numerical solution to equations of motion in FJNW

```
import numpy as np
from scipy.integrate import solve_ivp
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from matplotlib.patches import Wedge
import sympy as sp
#Setting up the constants
M = 1
n = 0.5
#Ultrastatic generic
reprad1 = []
reprad2 = []
reprad3 = []
def equations_of_motion(tau, current_state):
    global vtheta
    global reprad1, reprad2, reprad3
    #Step-wise assignment
    t, r,phi, vt, vr,vphi = current_state
    f = 1-(2*M)/(n*r)
    B = (2*q*np.sqrt(1-n**2))/(n*(q*np.sqrt(1-n**2)*np.log(f)+2))
    if(tau<0.01):
        #Setting initial value for vtheta to comply with
           normalization condition
        vtheta = np.sqrt(-1+(f**n)*(vt**2) - (f**(-n))*(vr**2) -
           (r**2)*(f**(1-n))*(vphi**2))/np.sqrt(r**2*f**(1-n))
    #Second derivative of t wrt tau
    d2t_dtau2 = -(2*M*vt*vr*(1+B/2))/(f*r**2)
    #Second derivative of r wrt tau
    d2r_dtau2 = -(f**(n-1)*M*(f**n*vt**2+B))/(r**2)+(M*vr**2*(1-B
       ))/(f*r**2) - (M*(1+n)/n- r)*vphi**2
    #Second derivative of phi wrt tau
    d2rphi_dtau2 = (2*M*vr*vphi)*(B/2 - 1+ r/M -1/n)/(f*r**2)
    #Sanity check for the norm of the four-velocity - diverges
       very quickly
    norm = -f**n*vt**2 + f**(-n)*vr**2 + r**2*f**(1-n)*vphi**2 +
       r**2*f**(1-n)*vtheta**2
    print("Norm is: ", norm)
    #Adding values to the repulsive radius if second derivative
       is positive
    if(q<0 and d2r_dtau2>0):
        reprad1.append(r)
    if(q==0 and d2r_dtau2>0):
        reprad2.append(r)
    if(q>0 and d2r_dtau2>0):
        reprad3.append(r)
    return [vt, vr,vphi, d2t_dtau2, d2r_dtau2,d2rphi_dtau2]
#Initial conditions, correspond to t,r,phi, dot(t), dot(r), dot(
   phi)
#Not dependent on t, phi
initial_state = [0, 1e1, 0, 1e4, -1e4,1e3]
```

```
tau_span = [0, 5e-2]
tau_eval = np.linspace(*tau_span, int(1e4))
#Particle charge
q_span = [-3e0,0,3e0]
r_init = initial_state[1]
phi_init = initial_state[2]
vr_init = initial_state[4]
vphi_init = initial_state[5]
solutions = []
for charge in q_span:
    q = charge
    #Solution calculated by solve_ivp - lower setting of rtol and
        atol has no further effect
    solution = solve_ivp(equations_of_motion, tau_span,
       initial_state,
                         t_eval=tau_eval, method='RK45', rtol=1e-10,
                            atol=1e-10)
    #r and phi assignment
    solutions.append([solution.y[1], solution.y[2]])
fig, axes = plt.subplots(1, 3, figsize=(12, 4))
iter = 0
for ax, sol in zip(axes, solutions):
    #Polar to Cartesian coordinates
    x = np.cos(sol[1])*sol[0]
    y = np.sin(sol[1])*sol[0]
    #Defining the axis limits so the motion is visible
    axlimit = 20
    ax.set_xlim(-axlimit,axlimit)
    ax.set_ylim(-axlimit,axlimit)
    #Curvature singularity at a point of 2M/n
    circle = patches.Circle((0, 0), 2*M/n, color='grey', fill=
       True, linewidth=1, linestyle = '--', label = 'Curvature
       singularity')
    ax.add_patch(circle)
    #Adding the repulsive sphere to the respective charge if it
       exists
    if(iter==0 and reprad1):
        ring = Wedge(center=(0, 0),
            r=max(reprad1),
            theta1=0,
            theta2=360,
            width=max(reprad1)-min(reprad1),
            facecolor='green',
            alpha = 0.3
            , label = 'Repulsive radius')
        ax.add_patch(ring)
    if(iter==1 and reprad2):
        ring = Wedge(center=(0, 0),
            r=max(reprad2),
            theta1=0,
            theta2=360,
            width=max(reprad2)-min(reprad2),
            facecolor='green',
            alpha = 0.3
            , label = 'Repulsive radius')
        ax.add_patch(ring)
    if(iter==2 and reprad3):
        ring = Wedge(center=(0, 0),
```

```
                r=max(reprad3),
                theta1=0,
                theta2=360,
                width=max(reprad3)-min(reprad3),
                facecolor='green',
                alpha = 0.3
                , label = 'Repulsive radius')
            ax.add_patch(ring)
            print(max(reprad3), min(reprad3))
        cur_q = q_span[iter]
        if(iter==2):
            #Charge singularity is at the sphere of radius calculated
                here (only for positive q)
            r_c, M_c, n_c, q_c = sp.symbols('r M n q', real=True,
               positive=True)
            sqrt_term = sp.sqrt(1 - n_c**2)
            f_c = 1 - (2*M_c)/(n_c*r_c)
            denominator = q_c*sqrt_term * sp.log(f_c) + 2
            solution = sp.solve(denominator, r_c)
            numerical_solution = [expr.evalf(subs={M_c: M, n_c: n,
               q_c: cur_q}) for expr in solution]
            circle = patches.Circle((0, 0), numerical_solution[0],
               color='blue', fill=False, linewidth=1, linestyle = '--
               ', label = 'Charge singularity')
            ax.add_patch(circle)
        ax.set_xlabel('x')
        ax.set_ylabel('y')
        ax.scatter(x, y, s=0.3)
        #Scaling the arrow
        arrow_factor = 3e-4
        #Arrow in cartesian coordinates for the initial velocity
        ax.quiver(r_init*np.cos(phi_init), r_init*np.sin(phi_init),
                (vr_init*np.cos(phi_init) - r_init*np.sin(phi_init)*
                   vphi_init)*arrow_factor
                ,(vr_init*np.sin(phi_init) + r_init*np.cos(phi_init)*
                   vphi_init)*arrow_factor,
            angles='xy', scale_units='xy', scale=0.7, color = 'violet
               ', alpha = 0.65)
        iter+=1
        ax.set_title(f"q' = {cur_q}")
        ax.legend(loc='lower right', labelspacing=0.1)
plt.tight_layout()
plt.savefig('polar_fjnw.png', dpi=300, bbox_inches='tight')
plt.show()
```

A.2 Effective potential in FJNW spacetime

```
import numpy as np
import matplotlib.pyplot as plt
import sympy as sp
def charge_singularity(Mm, Nn, Qq):
    #Calculates the symbolic charge singularity, then returns the
        values for our system
    r, M, n, q = sp.symbols('r M n q', real=True, positive=True)
    f = 1-(2*M)/(n*r)
    denominator = q*sp.sqrt(1-n**2)*sp.log(f)+2
```

```
    solution = sp.solve(denominator, r)
    numerical_solution = [expr.evalf(subs={M: Mm, n: Nn, q: Qq})
        for expr in solution]
    return numerical_solution
m = 1e-4
M = 1e0
n = 0.5
E = 1e-1
L = 5e0
r = np.linspace(2*M/n - 0.03, 30, 10000)
f = 1-(2*M)/(n*r)
multiply_factor = True
phi = (np.log(f))*np.sqrt(1-n**2)/2
q_span = [-3e0,0,3e0]
fig, ax = plt.subplots(1, 3,figsize=(18,6))
for iter,q in enumerate(q_span):
    denominator = (m*(1+q*phi))**2
    V_eff = (L**2*f**(2*n-1))/(denominator*(r**2)) + f**n
    Efactor= E/denominator
    if(multiply_factor):
        #We multiply by the factor and have energy constant
        V_eff*=denominator
        Efactor*=denominator
        ax[iter].plot(r, V_eff, label=r'$V_{\mathrm{eff}}(r)$',
            linewidth = 2)
        ax[iter].plot(r, Efactor , label=r'$E$')
    else:
        #Plotting in logarithmic scales
        ax[iter].plot(r, np.log(V_eff), label=r'$\ln{V_{\mathrm{
            eff}}(r)}$', linewidth = 2)
        ax[iter].plot(r, np.log(Efactor) , label=r'$\ln{E_{mod}(r
            )}$')
    ax[iter].set_title(f"q =  {q}")
    maxsize = max(V_eff[~np.isnan(V_eff)])
    minsize = min(V_eff[~np.isnan(V_eff)])
    ax[iter].set_ylim(10, maxsize +1 if maxsize<50 else 25)
    ax[iter].set_xlim(2*M/n-1,30)
    if(iter==2):
        #Calculation of the charge singularity
        ax[iter].axvline(x=charge_singularity(M, n, q), color='
            blue', linestyle='--', linewidth=2, alpha = 0.7, label
             = 'charge singularity')
    ax[iter].axvline(x=2*M/n, color='red', linestyle='--',
        linewidth=2, alpha = 0.7, label = 'curvature singularity')
    ax[iter].set_xlabel('r')
    ax[iter].legend(loc='upper right')

plt.tight_layout()
plt.savefig('effective_fjnw.png', dpi=300, bbox_inches='tight')
plt.show()
```